MME 351 – MARINE ELECTRICAL TECHNOLOGY

CONTENT:

Power generation: Selection of DC and AC generators, Merits and Demerits of AC and DC. Construction & Characteristics of brushless high speed AC generators with automatic Voltage regulator. Rules and Regulations governing electrical machinery on ships, location and installation of generators sets. Rating and characteristics of main switch boards. Arrangements of preferential trips and protective devices with wiring diagrams.

Ship power distribution system: regulations regarding layout of distribution system on general cargo ships and on oil tanker navigation lights and running light indicator panel. Special requirements for lighting of deck and pump house of oil tankers. Wiring appliances. Determination of wire size for AC power circuits.

System earthing: Isolated neutral systems, resistance earthed system.

Motor and Control Equipment: Special requirements of motors and starters for anchor Windlass & Capstan, wiring diagram, lowering & hoisting arrangement, overload protection, magnetic disc, brakes. Centralised control of motors in machinery spaces. Classes of insulation (A, B and E) and Temperature rise.

Miscellaneous electrical equipment: Sequential starting and cut outs for an automatic fired boiler incorporating safety devices & combustion control equipment. Sequential starting for refrigerating plants, incorporating safety devices like high pressure cut out, cooling water failure cut out, automatic defrosting with help of timer.

Reference materials

1. Patel, M.R., Shipboard Propulsion, Power Electronics, and Ocean Energy, CRC Press, 1st ed, 2012.
2. Patel, M.R., Shipboard electrical power systems, CRC Press, 1st ed, 2011.
3. Bird, J., Electrical Circuit Theory and Technology, Elsevier, 4th ed, 2010.

UNIT 1

POWER GENERATION IN MARINE VESSELS

1.1 **Introduction**

All marine vessels require electricity for various applications. Electrical power is used by the following components/plants and systems, among others.

(a). Propulsion system
(b). Steering gear system
(c). Emergency bilge pump
(d). Watertight doors
(e). Firefighting system
(f). Ship's navigation lights and emergency lights
(g). General lighting
(h). Communication and alarm system
(i). Motors
(j). Refrigeration

Electricity is produced using machines called generators. A generator is a machine that converts mechanical energy into electrical energy. It works on the principle of Faraday's Law of Electromagnetic Induction. Faraday's law states that electromotive force -EMF or voltage – is generated in a current-carrying conductor that cuts a uniform magnetic field. This can either be achieved by rotating a conducting coil in a static magnetic field, or by rotating the magnetic field that contains the stationary conductor. The preferred arrangement is to keep the coil stationary because it is easier to draw induced alternating current from a stationary armature coil than a rotating coil. There are two kinds of generators. These are AC generators and DC generators.

1.2 **AC GENERATORS**

AC generators produce voltages that are time varying. The voltage waveforms are sinusoidal at frequencies or either 50Hz or 60Hz. In principle, any AC generator can be called an alternator. However, the term 'alternator' is usually used to refer to small rotating machines driven by automotive and other internal combustion engines. An alternator that uses a permanent magnet for its magnetic field is called a magneto. Alternators in power stations driven by steam turbines are called turbo-alternators.

1.2.1 **Major parts of AC Generators**

Figure 1.1 shows the major constructional features of a generator. The major components AC generators are:

(a) Field
(b) Armature
(c) Prime Mover
(d) Rotor
(e) Stator
(f) Yoke
(g) Slip Rings

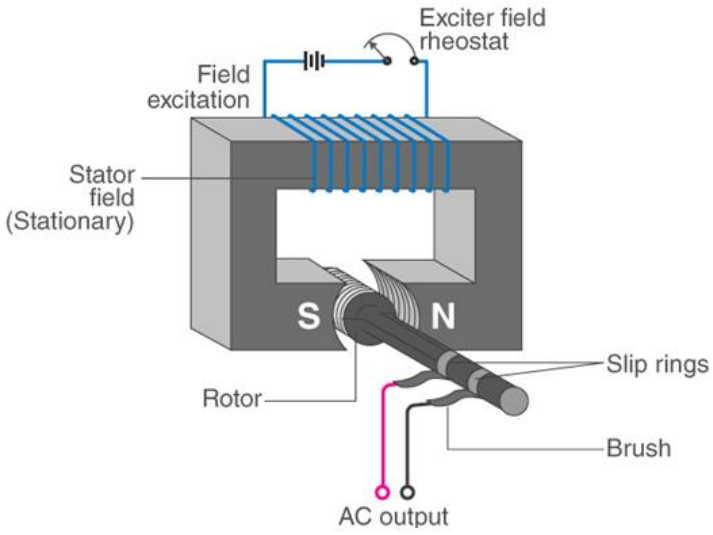


Fig. 1.1: Major parts of an AC generator

(a) ***Field***

The field consists of coils of conductors that receive voltage from a dc source and produce magnetic flux. The magnetic flux in the field cuts the armature to produce voltage. This voltage is the output voltage of the AC generator.

(b) ***Armature***

The part of an AC generator in which the voltage is produced is known as an armature. This component primarily consists of coils of wire that are large enough to carry the full-load current of the generator.

(c) ***Rotor***

The rotating component of the generator is known as a rotor. The generator's prime mover drives the rotor. Based on the type of generator, this component may either be the armature or the field. The rotor will be the armature if the voltage output is generated there; the rotor will be the field if the field excitation is applied there. Figure 1.2 shows images of rotors of generators.

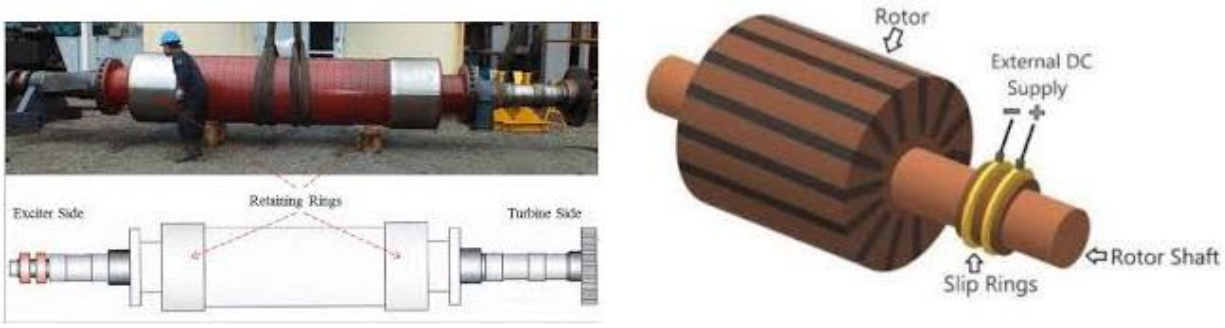


Fig 1.2 Rotor of generator

(d) ***Prime mover***

The component used to drive an AC generator is known as a prime mover. The commonly used prime movers are diesel engines, steam engines, steam turbines, gas turbines and water turbines.

(e) ***Stator***

The stator of an AC generator is the stationary part. Like the rotor, this component may house the armature or the field, depending on the type of generator. The stator will be the armature if the voltage output is generated there while it will be the field if the field excitation is applied there.



Fig. 1.3: Stator of generator

(f) **Yoke**

The yoke performs the following functions:

(a) It saves the purpose of outermost cover of the dc machine so that the insulating materials get protected from harmful atmospheric elements like moisture, dust and various gases like SO_2, acidic fumes etc.

(b) It provides mechanical support to the poles.

(c) It forms a part of the magnetic circuit. It provides a path of low reluctance for magnetic flux.

To provide low reluctance path, it must be made up of some magnetic material. It is prepared by using cast iron because it is the cheapest. For large machines rolled steel or cast steel, is used which provides high permeability i.e., low reluctance and gives good mechanical strength.

(g) ***Slip rings***

Slip rings are electrical connections that are used to transfer power to the rotor or from the rotor of an AC generator. They are typically designed to conduct current from a stationary device to a rotating one. Fig. 1.4 is an image of slip rings.

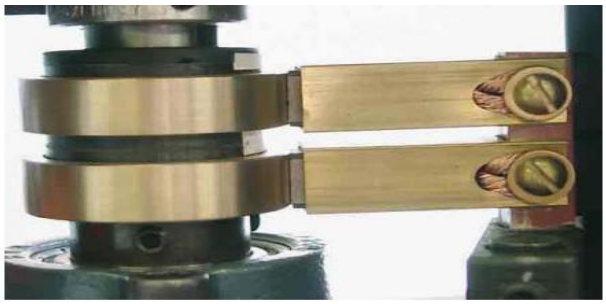


Fig. 1.4: Slip rings

1.2.2 **Working principle**

A generating action requires the following basic components to exist:

(a) Conductor or a coil
(b) Flux
(c) Relative motion between the conductor and the flux

For reasons of cost and simplicity, most alternators use a rotating magnetic field with a stationary armature. To have a large voltage as output, several conductors are connected in a special arrangement to form a winding. Typically, a rotor having field windings turns (aided by a mechanical input) within a stator that has armature windings. The magnetic field set up in the field windings cuts the armature conductors, generating an induced EMF (electromotive force). The rotating magnetic field induces an AC voltage in the stator windings.

1.3 **DC GENERATORS**

The generator is provided with a magnetic field by sending dc current through the field coils mounted on laminated iron poles and through armature winding. A short air gap separates the surface of the rotating armature from the stationary pole surface. The magnetic flux coming out of one or more worth poles crossing the air gap, passes through the armature near the gap into one or more adjacent south poles. The direct current leaves the generator at the positive brush, passes through the circuit, and returns to the negative brush. The terminal voltage of a dc generator may be increased by increasing the current in the field coil and may be reduced by decreasing the current. Generators are generally run at practically constant speed by their prime mores.

1.3.1 **Parts of a DC Generator**:

The DC generator has the following major parts.

(a) Yoke

(b) Magnetic Poles: Pole core, pole shoe
(c) Field
(d) Armature Core
(e) Armature
(f) Commutator
(g) Brushes

The stationary parts and rotating parts are separated from each other by an air gap. The stationary part of a D. C. machine consists of main poles, designed to create the magnetic flux, commutating poles interposed between the main poles and designed to ensure sparkless operation of the brushes at the commutator and a frame/yoke. The armature is a cylindrical body rotating in the space between the poles and comprising a slotted armature core, a winding inserted in the armature core slots, a commutator and brush.

(a) **Magnetic Poles**

Each pole is divided into two parts as shown in figure 1.5. These are: (a) pole core and (b) pole shoe.

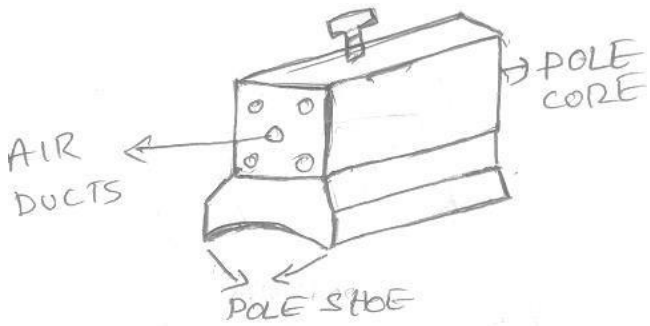


Fig. 1.5: Pole core and pole shoe

Functions:

1. Pole core basically carries a field winding which is necessary to produce the flux.
2. It directs the flux produced through air gap to armature core to the next pole.
3. Pole shoe enlarges the area of armature core to come across the flux, which is necessary to produce larger induced emf.

To achieve this, pole core has been given a particular shape.

Choice of material: It is made up of magnetic material like cast iron or cast steel. As it requires a definite shape and size, laminated construction is used. The laminations of required size and shape are stamped together to get a pole which is then bolted to yoke.

(b) **Armature**

It is further divided into two parts namely, (a) Armature core and (b) Armature winding.

Armature core is cylindrical in shape mounted on the shaft. It consists of slots on its periphery and the air ducts to permit the air flow through armature which serves cooling purpose. Figure 1.6 depicts this.

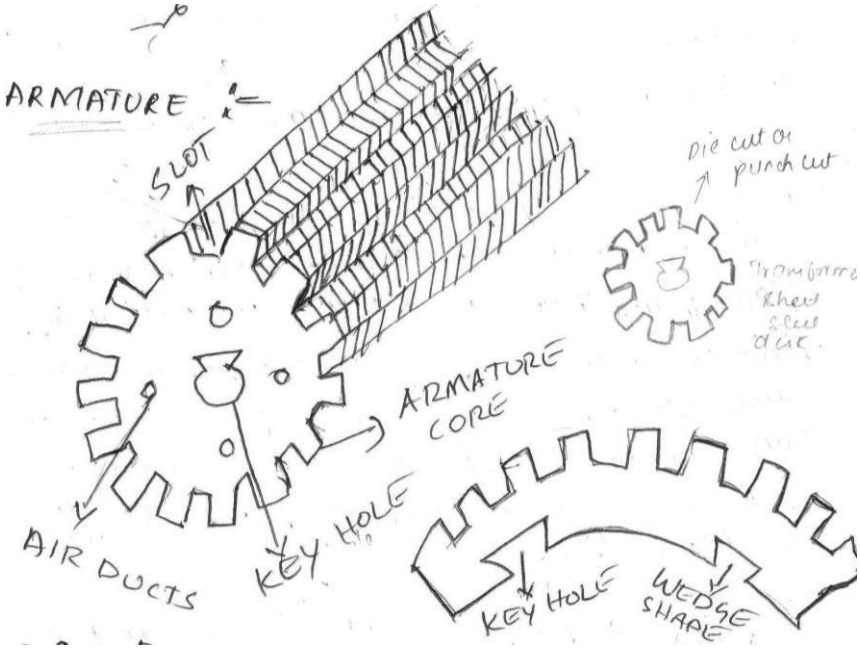


Fig. 1.6: Armature core

Functions:

1. Armature core provides house for armature winding i.e., armature conductors.
2. To provide a path of low reluctance path to the flux it is made up of magnetic material like cast iron or cast steel.

Choice of material: As it has to provide a low reluctance path to the flux, it is made up of magnetic material like cast iron or cast steel. It is made up of laminated construction to keep eddy current loss as low as possible.

(c) **Armature winding**

Armature winding is nothing but the inter connection of the armature conductors, placed in the slots provided on the armature core. When the armature is rotated, magnetic flux gets cut by armature conductors and emf gets induced in them.

Function

1. Generation of emf takes place in the armature winding in case of generators.
2. To carry the current supplied in case of dc motors.
3. To do the useful work it the external circuit.

Choice of material: As armature winding carries entire current which depends on external load, it must be made up of conducting material, which is copper.

(d) **Field winding**

The field winding is wound on the pole core with a definite direction. Figure 1.7 shows laminated core with field winding.

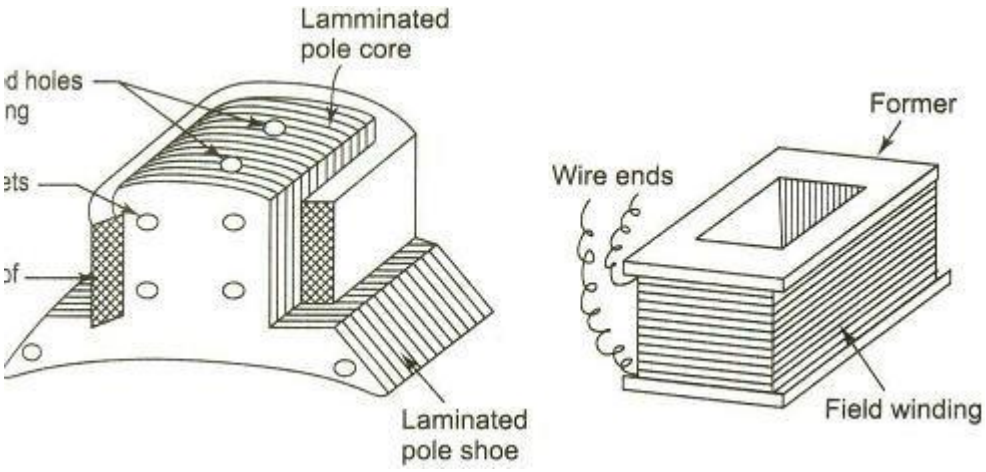


Fig. 1.7: Laminated pole with field winding

Function:

Field windings carry currents that excite pole cores to produce magnetic fields. Field winding is divided into various coils called field coils. These are connected in series with each other and wound in such a direction around pole cores such that alternate N and S poles are formed.

(e) **Commutator**

Rectification of generated ac voltage into dc voltage is done in dc generators using commutators. Fig. 1.8 shows a commutator.

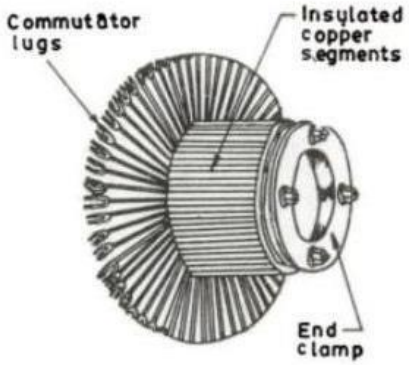


Fig. 1.8 Commutator of dc generator

Functions

1. To facilitate the collection of current from the armature conductors.
2. To convert internally developed alternating emf (AC) to DC.

3. To produce unidirectional torque in case of motor.

*Choice of material***:** As it collects current from armature, it is also made up of copper segments. It is cylindrical in shape and is made up of wedge-shaped segments which are insulated from each other by thin layer of mica.

(f) **Brushes**

Brushes are stationary and rest on the surfaces of commutators. Brushes are rectangular in shape. They are housed in brush holders, which are usually of box type. The brushes are made to press on the commutator surface by means of a spring, whose tension can be adjusted with the help of lever. A flexible copper conductor called pigtail is used to connect the brush to the external circuit. Figure 1.9 shows details of a brush.

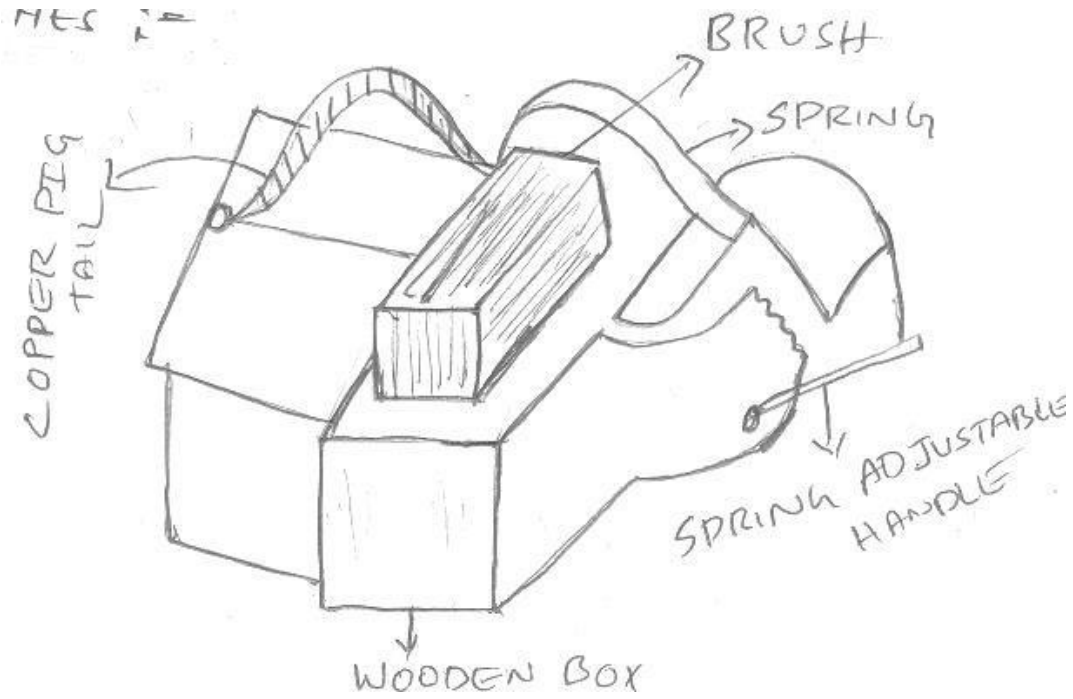


Fig. 1.9: Brush of DC generator

Functions: To collect current from commutator and make it available to the stationary external circuit.

Choice of material: Brushes are normally made up of soft material like carbon.

1.4 **Comparison between AC and DC generators**

Table 1.1 provides a comparison between AC and DC generators.

Table 1.1: Comparison between AC and DC generators

No.	Differentiating property	AC generator	DC generator
1	*Definition*	AC generator is a mechanical device that converts mechanical energy into AC electrical power.	DC generator is a mechanical device that converts mechanical energy into DC electrical power.
2	*Direction of Current*	In an AC generator, the electrical current reverses direction periodically.	In a DC generator, the electrical current flows only in one direction.
3	*Basic Design*	In an AC generator, the coil through which the current flows is fixed while the magnet moves. The construction is simple, and costs are reduced.	In a DC generator, the coil through which the current flows rotate in a fixed field. The overall design is simple, but construction is complex due to commutators.
4	*Commutators*	AC generators do not have commutators.	DC generators have commutators to make the current flow in one direction only.
5	*Rings*	AC generators have slip-rings.	DC generators have commutators.
6	*Efficiency of Brushes*	Since slip-rings have a smooth and uninterrupted surface, they do not wear quickly and are highly efficient.	Both brushes and commutators of a DC generator wear out quickly and thus are less efficient.
7	*Short Circuit Possibility*	As the brushes have high efficiency, a short circuit is very unlikely.	Since the brushes and commutators wear out quickly, sparking and short circuit possibility is high.
8	*Rotating Parts*	The rotating part in an AC Generator is a low current high resistivity rotor.	The rotating part in a DC generator is generally heavy.
9	*Current Induction*	In an AC generator, the output current can be either induced in the stator or in the rotor.	In a DC generator, the output current can only be induced in the rotor.
10	*Output Voltage*	AC generators produce a high voltage which varies in amplitude and time. The output frequency varies (mostly 50Hz to 60Hz).	DC generators produce a low voltage when compared to AC generator which is constant in amplitude and time i.e. output frequency is zero.

11	*Maintenance*	AC generators require little maintenance and are highly reliable.	DC generators require frequent maintenance and are less reliable.
12	*Types*	AC generators are or varied types such as 3-Phase generators, single-phase generators, synchronous generator, induction generator, etc.	DC generators are mainly two types which are Separately excited DC generator and Self-excited DC generator. According to field and armature connection, they can be further classified as DC series, shunt, or compound generators respectively.
13	*Cost*	The initial cost of an AC generator is high.	The initial cost of a DC generator is low.
14	*Distribution and Transmission*	The output from AC generators is easy to distribute using a transformer.	The output from DC generators is difficult to distribute as transformers cannot be used.
15	*Efficiency*	AC generators are very efficient as the energy losses are less.	DC generators are less efficient due to sparking and other losses like copper, eddy current, mechanical, and hysteresis losses.

1.5 **GENERATOR LOCATION**

A solid, level mounting platform is particularly important for the proper operation of a generator. A location that will allow adequate space on all sides for ventilation and servicing, preferably on, and parallel with, the keel or vessel center line is recommended. Generators must be kept away from living quarters, and away from bilge splash and vapours. Enough room must be created for access to the following service points:

- Negative and positive battery connections
- Fuel inlet and outlet connections
- Oil fill and oil drain hose
- Raw water inlet and outlet
- Location of fuel/water filters.

A generator will rock on its mounts in rough seas, therefore clearance is necessary on all sides. Larger generators require up to a three-inch (76mm) clearance. Generator should not be supported from bulkheads since they can create drumming effects and transmit noise and vibration.

Generator location is preferable in the same room or compartment as the propulsion engine, as this is usually a well-ventilated area, insulated, close to the fuel supply and the center of the electrical load distribution. However, a generator cannot be installed in the propulsion engine compartment unless specific conditions are met. For example, USCG regulation 33CFR183 which pertains to

gasoline fuel systems, requires a generator operating in a gasoline fuel environment to be "ignition protected."

It is critical to note that gasoline fire or explosion can result in severe personal injury or death. Therefore, a gasoline generator should not be installed in the same room or compartment as a diesel engine or generator. The diesel unit may not be ignition protected and may ignite gasoline fumes. A gasoline generator must be installed in its own room or compartment on a diesel-powered boat. Also, a diesel generator should not be installed in the same room or compartment as a gasoline engine.

1.6 **Emergency Power**

In case of the failure of the main power generation system on a ship, an emergency power system or a standby system is also present. The emergency power supply ensures that essential machinery and systems are powered. Essential machinery and systems in ships include the following:

(a) Steering gear system
(b) Emergency bilge
(c) Watertight doors.
(d) Fire fighting system.
(e) Ships navigation lights and emergency lights.
(f) Communication and alarm system.

Emergency power can be supplied by batteries or emergency generators, or both. The rating of an emergency power plant should be such that it provides the needed supply to all essential machinery and systems.

Emergency generator is normally located outside the machinery space of a ship. This is done mainly to provide unimpeded access to the engine room during emergencies. A switch board in the emergency generator room supplies power to different essential machinery.

1.7 **SHORE POWER**

When a ship docks, it no longer needs energy for propulsion. However, ships may still be large consumers of energy when stationary as several of the ship functions such as ventilation/heating/cooling, pumps, control systems and cargo handling systems still operate. Consequently, if the ship’s generator continue to run when in port, there will be local noise and emissions. Rather than letting the generators on board produce the needed electricity, electricity can be tapped from the power grid of the land. Power supplied to ships from land is called shore power. Figure 1.10 depicts the supply of shore power to a ship. Shore power is also referred to as cold ironing, shore-to-ship power or alternative maritime power.

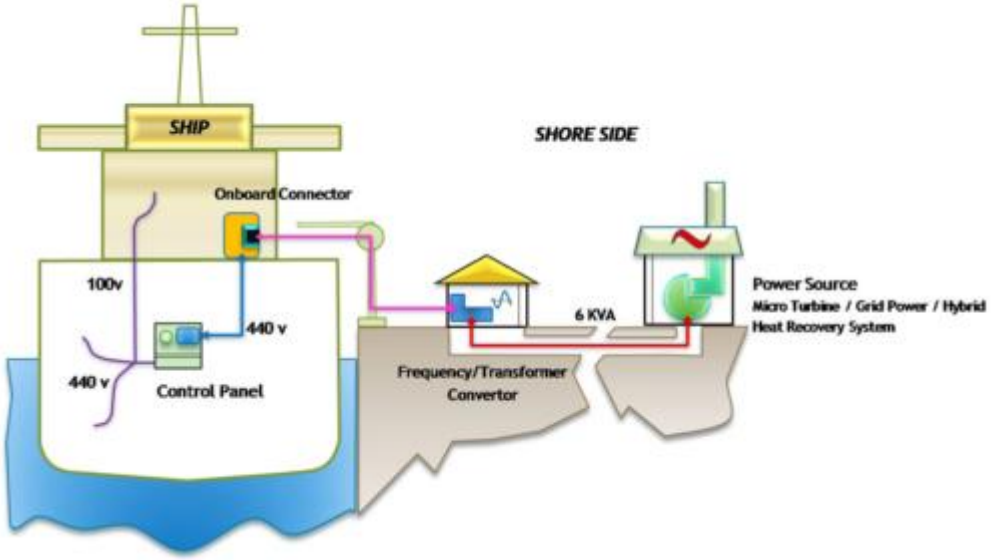


Fig. 1.10: Shore power

Shore power can be installed for all types of vessels for that require power in harbour. Shore power has been used extensively for many years for vessels with moderate power requirements; typically less than 50 to 100 kW. These vessels are capable of making use of normal grid voltage and frequency.

For large vessels with higher power requirements (100 kW up to 15 MW), it gets a bit more complicated. To serve these vessels with shore power, dedicated and relatively costly installations are required, both on land and on board the vessels. This may include upgrading the grid capacity, frequency converters and complex high power connectors. Consequently, relatively few vessels and ports are capable of making use of shore power, even though the environmental upsides are considerable.

Shore power may potentially eliminate the local noise and air pollution related to ship activity in a port. Depending on the energy source, it may also contribute positively to the climate driving effects of ship operation, but as an isolated initiative, it is generally not considered to be among the most cost effective climate initiatives.

The grid power solution and the frequency converters typically represent the costliest elements on the shore side. Depending on the availability of grid power and the power requirements, the cost of installing shore power on the shore side will vary considerably. On the land side, the high power cold ironing system consists of the following:

(a) High voltage grid to the port
(b) Frequency and voltage convertors/transformers
(c) Control panels and connection boxes
(d) Cable reel and connectors

On the ship side the following will have to be installed:

(a) Transformer
(b) Power distribution system
(c) Control panel
(d) Frequency converter (optional for greater flexibility)
(e) Connectors and cable reel (optional for greater flexibility)

UNIT 2

POWER DISTRIBUTION ON SHIP

2.1 INTRODUCTION

Power generated on a ship must be distributed to all electrical loads on board. The network that generates and distributes power to all needed parts or components of a ship is referred to as a distribution system. Figure 2.1 shows a typical distribution system for a ship.

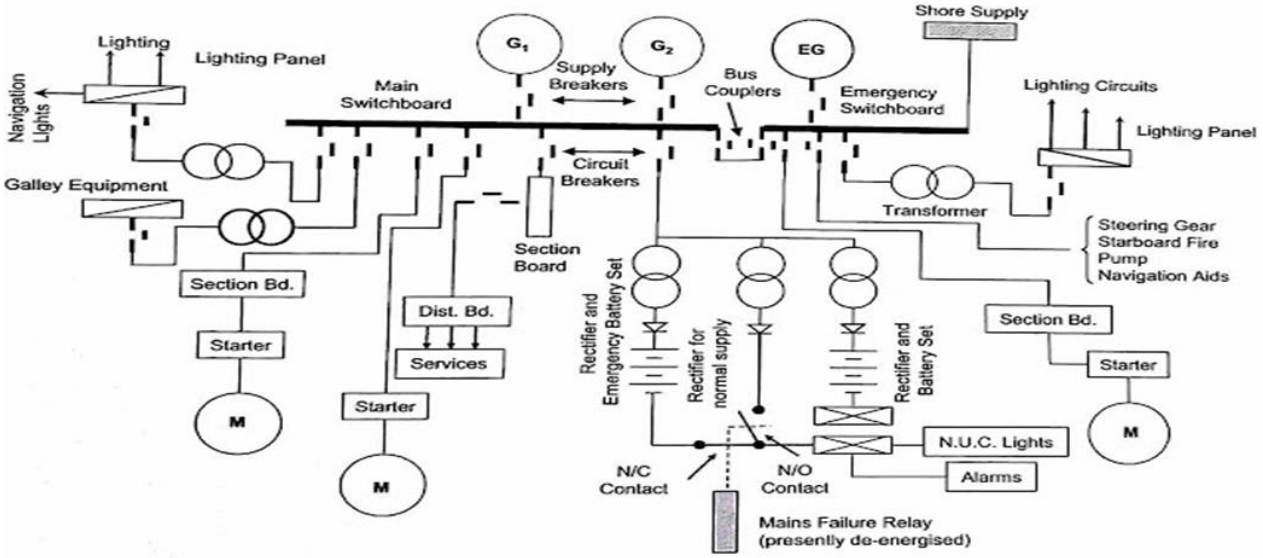


Fig. 2.1: Typical ships' distribution system

A ship's distribution system consists of different components for distribution and safe operation of the system. They are:

(a) Ship generator consisting of prime mover and alternator.
(b) Main switch board which is a metal enclosure taking power from the diesel generator and supplying it to different machinery.
(c) Bus Bars which act as a carrier and allow the transfer of power from one point to another.
(d) Circuit breakers which act as a switch and in unsafe condition can be tripped to avoid breakdown and accidents.
(e) Fuses as safety device for machinery.
(f) Transformers to step up or step down the voltage.

In a power distribution system, the voltage at which the system works is usually 440V or 450V. There are some large installations where the voltage is as high as 6600V. Power is supplied through circuit breakers to large auxiliary machinery at high voltage. For smaller supply, fuse and miniature circuit breakers are used.

2.2 **MAIN SWITCH BOARD**

The main switchboard is the main power distribution center of a ship. The main switch board has various measuring and monitoring devices like ammeters, voltmeters, frequency meters, watt meters, synchroscope, and power factor meters to provide details about power supply.

The main switch board gets its supply from the main generators located in the engine room. The distribution system can be divided into Feeder side and the Load side. The feeders are the generators and the loads are various ship board auxiliary machineries. The Load side is further divided into Essential and Non-Essential services. Essential services are critical auxiliaries which directly affect the safety of personnel, and ships navigation and propulsion systems. They include supply to navigational aids like radars, communication equipment, navigational lights, and steering gear motors. These essential services may be supplied directly from the main switch board or via sectional boards or distribution boards. Non-essential services are those that do not affect the safety of ship and personnel. For example, air conditioning compressors and fans are non-essential services as they don't affect the safety of ships or personnel. The operating voltage is stepped down to 220 volts and this voltage is given to the 220 Volts auxiliaries' panel. From this panel, the supply goes to various lighting systems and other sockets for accommodation and galley for domestic use.

The following should be considered in the installation of distribution switchboards:

1. Main Switch Board (MSB) should be installed on a raised platform.
2. The location where Switch Board or MSB is installed should have good ventilation, illumination and affected by minimum vibration.
3. No water or oil pipes should pass above the Switch Board or MSB.
4. There should be adequate space around allowing comfortable operation ans maintenance.
5. The operating floor space in front and behind the Switch Board or MSB should be covered with anti-skid, insulated rubber mats with a minimum thickness of 15 mm.
6. Panel doors should be in locked position.
7. Door opening mechanism should be interlocked with power supply.
8. The enclosure should be drip proof.
9. MSB should be dead front type. (i.e. All the live conductor should pass behind MSB).
10. Every equipment onboard must be properly earthed. Also, MSB must be properly earthed.
11. Every incoming and outgoing feed must be protected by a switch fuse or circuit breaker.
12. Insulation resistance should not be less than 1 Mega Ohm in any case. Insulation Resistance (IR) to be checked and records are maintained.
13. All meters must be calibrated periodically, and records are maintained.
14. All the meters, switches, fuses, push button, indicating lamps, etc. should be in working condition and identified with individual name plate or tag.
15. All the interlocks must be in working condition.
16. Operating switch gear, tools such as circuit breaker handle, fuse puller, lamp grip, etc. must be readily available.

17. Red tag / name plate indicating high voltage and danger sign should be fitted.
18. Appropriate ready to use fire extinguishers should be positioned nearby. (CO and DCP).
19. Charts indicating drill procedures in case of electric shocks must be displayed near MSB.
20. Wiring diagram laminated or framed indicating supply system must be displayed nearby.
21. All the appropriate spares should be properly preserved and protected, individually packed and identified, and should be readily available for use.
22. Earth fault indication should be fitted and displayed in front panel.
23. 24 Volt battery supply emergency light should be provided near MSB.

2.3 **THE MAIN BUSBAR**

The main bus bar can be found inside the main switch board. Onboard a merchant ship, the supply is usually 440 volts, 60 Hertz, however higher voltages of 6.6kV are possible on LNG tankers, special purpose ships, and passenger vessels having electrical propulsion. The rating of the main bus bar is decided by the type of ship, the critical machinery on board required for cargo operations, and the machinery required for normal sea service. If a ship is steam propelled, it might have steam turbine driven alternators and a diesel driven alternator as a back-up. If the propulsion is by electric motors, then the vessel might require huge diesel driven generators which usually operate at 6.6kV. The main bus bar has three heavy, thick bars of conductor (usually copper), running horizontally throughout the length of the main switch board. Each of the conductors is insulated between each other and wherever it is supported or clamped to the switchboard. Table 2.1 provided typical power requirement of various marine vessels

Table 2.1 Typical power requirement for vessels

No.	Vessel	Typical power requirement
1	A 10,000 DWT general cargo ship	1 MW
2	Tankers	1.5 to 5 MW
3	Containerships (10000 TEU) (3.3kV)	8 MW
4	A Modern LNG carrier (6.6kV)	12 MW

Thus, the main bus bars are contained within the switchboard feeding various shipboard auxiliaries. A ship may contain two or more generators connected to the main bus bar via the circuit breaker. Various protection schemes are applied to generators against abnormal or fault conditions. From the main bus bar, electrical power is supplied to various shipboard auxiliaries like pumps, blowers and compressors.

2.4 **EMERGENCY SWITCHBOARD**

The *emergency switchboard* is the emergency power distribution center. The emergency switchboard distributes 450V, 220V or 120V power to auxiliaries which are vital, under emergency

conditions such as essential lighting, safety and communication systems, and power necessary for dead ship starting. Under normal operation, power for the emergency switchboard is supplied by the main generators through a bus-tie. In case of power failure, loss of voltage at the emergency switchboard will cause the emergency generator to automatically start. Simultaneously, the bus-tie will open, disconnecting the main and the emergency switchboards. Then, the emergency generator breaker will close, energizing the emergency board and providing the needed emergency power.

The emergency switchboard is composed of the following sections:

1. A generator and bus transfer section that provides control for operation of the emergency generator and the bus transfer unit.
2. A distribution unit with a 450V, 60Hz, 3 phase bus provides power for emergency power loads, and a 450V/120V transformer bank, the secondary of which supplies a 120V, 3 phase, 60 Hz bus that provides power for emergency lighting and I.C. (Interior Communication) equipment.
3. A 120V DC panel for battery distribution. An automatic charging unit is provided for charging the ship's emergency power battery.

2.5 **CIRCUIT BREAKERS**

A circuit breaker is designed to protect an electrical apparatus from damage caused by overload or short-circuit. Unlike a fuse, a circuit breaker can be reset (either manually or automatically) to resume normal operation. Circuit breakers may be single-phase (single-pole) or three-phase (three-pole). A circuit breaker generally comprises poles (set of contacts-one fixed, one moveable), operating mechanism, dielectric and auxiliaries. The types of circuit breakers are: miniature circuit breaker (MCB), moulded case circuit breaker (MCCB), air-break circuit breaker, air blast circuit breaker, sulphur hexafluoride gas circuit breaker, oil circuit breaker and vacuum circuit breaker.

2.6 **FUSES**

A fuse acts as both a protective and a disconnecting device (it combines the functions of sensing, comparing and interrupting current). It is the simplest and cheapest form of protection against excessive current. A fuse consists of a metallic element which melts (ruptures) and becomes discontinuous at a relatively high current, thus preventing the further passage of current. It must however, permit the maximum load current to flow continuously without operating or deteriorating. Generally, a fuse has three parts namely: the fuse link (wire), fuse carrier and fuse base. Fuses may be grouped into low voltage fuses and high voltage fuses. Fuses have the following advantages and disadvantages:

Advantages

(a) Fuses are simple to use
(b) Fuses are direct acting
(c) Fuses are cheap
(d) They do not fail
(e) They are very fast at high currents.
(f) Their operating characteristics can be graded.

Disadvantages

(i) Fuses require replacement after every operation
(ii) Fuses offer poor protection against low fault currents
(iii) They are non-directional
(iv) They have limited voltage breaking capabilities

The types of fuses include, miniature fuses, semi-enclosed or rewritable fuses and high rapturing capacity fuses.

2.7 RULES AND REGULATIONS GOVERNING ELECTRICAL MACHINERY ON SHIPS

The purpose of regulations is to achieve a satisfactory level of electrical safety in maritime electrical installations and in the operation of electrical equipment connected to the installations. Electrical safety of personnel and machinery users are at the heart of such rules and regulations.

Electrical safety comprises dangers posed by electricity, such as danger arising from contact with electricity, fire, and safety factors associated with the use of electricity and electrical equipment. Electrical safety includes the following:

(a) protection against contact with energized parts,
(b) protection against arcing/overheating (personal injury, danger of fire, explosions),
(c) consequential injury,
(d) protection against electrostatic charges,
(e) protection against undesired chemical processes caused by electricity (fx in connections and when using a hull return system),
(f) protection against harmful effects from electromagnetic fields,
(g) electromagnetic compatibility,
(h) safe construction, installation and maintenance of electrical equipment,
(i) safe system design, construction, use and maintenance of maritime electrical installations,
(j) safe procedures and safety measures for work on or near maritime electrical installations,
(k) power supplied in a safe manner,
(l) high quality power supply,
(m) protection against lightning,
(n) protection to prevent power cuts causing danger to life or health.

National and internal agencies and authorities, such as the international convention for the Safety of Life at Sea (SOLAS), that regulate marine industries develop various rules and regulations to ensure the safe operation of electrical machinery in marine vessels.

UNIT 3
SYSTEM EARTHING

3.1 INTRODUCTION

Earthing (or grounding) is a deliberate connection between an electrical power grid and earth. Distribution systems are earthed to create a reference point for the system voltage, to facilitate the detection and discriminative isolation of faults involving contact to earth, and to limit over-voltages under transient conditions. A neutral earthing system is a system in which the neutral is connected to earth, either solidly, or through a resistance or reactance of value sufficient to materially reduce transients, and to give sufficient current for selective earth fault protection devices to operate. On board a metallic ship, Earth is the hull and superstructure.

Generally, there are five methods of neutral earthing. They are:

(a) Unearthed neutral system
(b) Solid neutral earthed system
(c) Resistance neutral earthing system
    (i) Low resistance earthing
    (ii) High resistance earthing
(d) Resonant neutral earthing
(e) Transformer earthing

These five methods can also be broadly classified as unearthed system (insulated earth system) and earthed system. In unearthed neutral system, there is no internal connection between conductors and earth while in earthed systems, the neutral is either connected to earth, through no deliberate impedance (solid earthing), or through a form of impedance.

On land, earth connection is ‘tied to’ the neutral connection typically at the local electrical substation, and at the point where it enters a building. If an earth fault occurs on a land-based installation, a protective device (eg. Residual current device, RCD) will trip quickly, and cut the supply. This is ideal for land use, as the priority is to protect human life and livestock. However, on the ship, the priority is to keep essential equipment running. For example, it would not be acceptable on a ship, if an electrical fault causes the steering gear to stop working, as this may be catastrophic.

3.2 EARTHED AND INSULATED NEUTRAL SYSTEMS

Onboard a ship, the priority requirement is to maintain continuity of the electrical supply to essential equipment in event of a single earth fault. The priority requirement ashore is the immediate isolation of earth-faulted equipment, which is automatically achieved in an earthed system.

To understand this better, a case study is used. If an earth fault occurs in an essential system like that of steering gear, then in case of insulated neutral system, no earth fault current will flow and the steering gear will continue to operate until there is a second earth fault in any equipment

onboard. Hence, even with a single earth fault, the essential service will continue to operate. On the other hand, if the system is earthed, then a single earth fault will cause heavy earth current to flow and operate the tripping mechanism, shutting down the steering gear. This will seriously compromise the safety of navigation.

Keeping this in mind, it is well understood that onboard a ship, a neutral insulated system is to be used. Shipboard main LV systems at 440V are normally provided with neutral insulated system. On the other hand, HV systems (1000V to 3.3kV) are usually provided with neutral earthing via a resistor. In HV systems, certain essential loads can be supplied by a transformer with its secondary insulated, to ensure no earth fault current flows in the equipment. This maintains the continuity of service.

The international convention for the Safety of Life at Sea (SOLAS) requires that the hazardous areas of tanker, viz: cargo area, pump room should have a neutral insulated system to prevent any stray earth current from flowing in the hull that could cause explosions. However, an exception is included in case a tanker has a 3.3kV system, the earthed system is permitted provided that the earthed system does not extend forward to engine room bulkhead and into hazardous areas.

Both the insulated neutral and earthed neutral system have got their own advantages and disadvantages. Just as it is easier in earthed neutral system to detect earth faults, it is easier in isolated neutral systems to maintain the continuity of service. However, an insulated system, is more effective than an earthed system, because the insulated system requires two earth faults on two different lines to cause the tripping of equipment.

The main advantages of insulated neutral system are:

(a) Continuity of service on occurrence of an earth fault.
(b) Earth fault currents can be kept low.

The principal disadvantages are:

(a) Occurrence of high transient over – voltages.
(b) Requirement of high level of insulation.
(c) Difficulty in detecting earth faults.

The main advantages of the earthed systems are:

(a) Equipment insulation does not require any special attention.
(b) Earth faults are detected automatically, and immediately isolated.
(c) Earth fault current flows for a short period of time thereby restricting damage.
(d) Arcing earth over-voltages can be avoided.

The main disadvantages are:

(a) Immediate disconnection and loss of service when an earth fault occurs.
(b) High earth fault currents flow, which can cause extensive damage and explosion.

Generally, in the marine world, the desire to maintain service amid an earth fault, and having low earth fault currents favours the insulated neutral system. The common practice, to derive benefits from both earthed and unearthed systems is to apply restricted earthed systems to specific services (e.g., domestic, hotel, galley), apply insulated systems to essential services, and split systems and apply redundancy. The following cases are common:

(a) **Low power -low voltage systems - solidly earthed**
Most low power, low voltage marine systems use a solidly earthed neutral system. This permits the use of the phase to neutral point as a source of low voltage for small power and lighting consumers. With this system a single earth fault will cause an overcurrent in the faulted circuit and this in turn will trip the circuit supply breaker to disconnect the service. For many services, this is quite acceptable, but for services such as steering gear, lubrication and fuel pumps, this could be an unacceptable practice.

(b) **Low voltage systems - insulated neutral**
Most low voltage marine systems (380 - 600 volts) have unearthed configurations. The main reason is that transient over-voltages on such low voltage systems are generally within equipment insulation level (under 2 kV). But more importantly, an unearthed system can still operate even if there is an earth fault present. With unearthed systems, however, it is important to have an earth fault detection system in place and to identify clear earth faults as quickly as possible. It should be remembered that when the first fault occurs on an insulated neutral system, the system immediately becomes "earthed" and consequently the next fault that occurs will be presented with a much lower zero sequence reactance. This can cause fault currents in excess of the 3 phase fault current for which most switchgear is rated. Consequently, to avoid the risk of protection equipment failure, any fault should be cleared as rapidly as possible. A prudent electrical engineer/Technician will check with the navigating staff to find a time slot when earth faulted equipment, can be worked on, without interrupting essential services, and brought back into full reliable service.

For systems in hazardous areas, insulated neutral systems should be used. Although an earthed neutral system will provide rapid fault clearance, the risk of a faulted cable causing flashover to earth, with resulting explosion, is high.

(c) **Medium voltage systems - high impedance earthed**
The high impedance earthing system offers several advantages such as: (i) low earth fault current which limits damage and reduces the fire risk, (ii) minimal earth fault flash hazard due to system overvoltages, and (iii) low protection equipment costs.

A variety of earthing methods are in use on medium voltage systems. On some UK designed tankers, insulated neutral systems are used at 3.3kV. Other systems use solidly earthed neutrals. Insulated neutral systems have certain inherent disadvantages and unless it is vital that particular equipment remains operational, even though a earth fault exists, it is recommended that insulated neutral systems be avoided. Although the earth fault current is inherently small, the integrity of the

system is at risk unless rapid fault detection and clearance is carried out. Furthermore, all equipment must be capable of operating with one phase at earth potential. For high voltage equipment this may require additional insulation. By solidly earthing the neutral, the earth fault current will undoubtedly trip the protection equipment and clear the fault. However, if the zero-sequence impedance, which limits the earth fault current, is considerably less than the positive sequence impedance, the high earth fault current may exceed that due to a three phase fault. This should be checked when determining protection equipment short-circuit current ratings.

Most medium voltage systems use resistance earthing. The resistance is connected between the neutral point and the ship's hull. The resistance limits the earth fault current to a low value, but one that is high enough to ensure selective operation of earth fault protective devices. The practice in the US is to limit the earth fault current to 3 times the system capacitive charging current (I_{co}). As a guide 6kV marine systems generally have a capacitance of about 1.4 mF. This means that the resistance should be sized to limit the earth fault current to about 5.5A (3 x I_{co}). This will then prevent any charge being built up on the system due to transient overvoltages. However, marine systems generally have high losses and consequent heavy damping; therefore, such over-voltages are generally not a problem. It is generally better to limit the earth fault current to a value just higher than $3I_{co}$ to ensure the operation of the earth current detection and protection equipment.

Although operation of a high voltage system with an earth fault is possible, it cannot be recommended and steps should always be taken to initially limit the fault current to as low a value as is acceptable, and secondly to isolate the equipment and repair the earth fault as soon as possible. If this is not done there is always a danger that the fault will escalate to a phase-to-phase fault and cause fire or extensive equipment damage.

To protect persons against the dangers of electric shock and possible fire hazard, metal enclosures, and other non-current-carrying metal parts of equipment must be earthed. This enables the potential at the time of earth fault, to become zero and provide the least resistance path for current to flow to earth (Ship’s hull).

3.3 CAUSES AND PREVENTION OF EARTH FAULTS

Earth faults arise due to

(a) dampness,
(b) mechanical damage,
(c) contamination due to dirt,
(d) temperature rise, and
(e) ageing.

Earth faults can be prevented by

(a) using appropriate enclosures to prevent the ingress of moisture and dust,
(b) offering protection against mechanical damage,
(c) using exact cable glands and seals on cable ends,

(d) regularly performing insulation resistance tests, and
(e) strictly maintaining equipment.

3.4 DETECTION OF EARTH FAULTS

Regulations require that earth fault indicators are fitted to main switch boards to indicate the presence of an earth fault on each isolated section of a distribution system. Earth fault indicators can be a set of lamps or an ohm meter calibrated in kΩ to show the system insulation resistance value to earth. The detection of single earth faults will permit early rectification of same to prevent the effect of a second earth fault occurring on another phase (or conductor) which will lead to dangerous consequences. Consequently, there are techniques to detect single earth faults. Figure 3.1 shows a schematic of an earth fault detection technique.

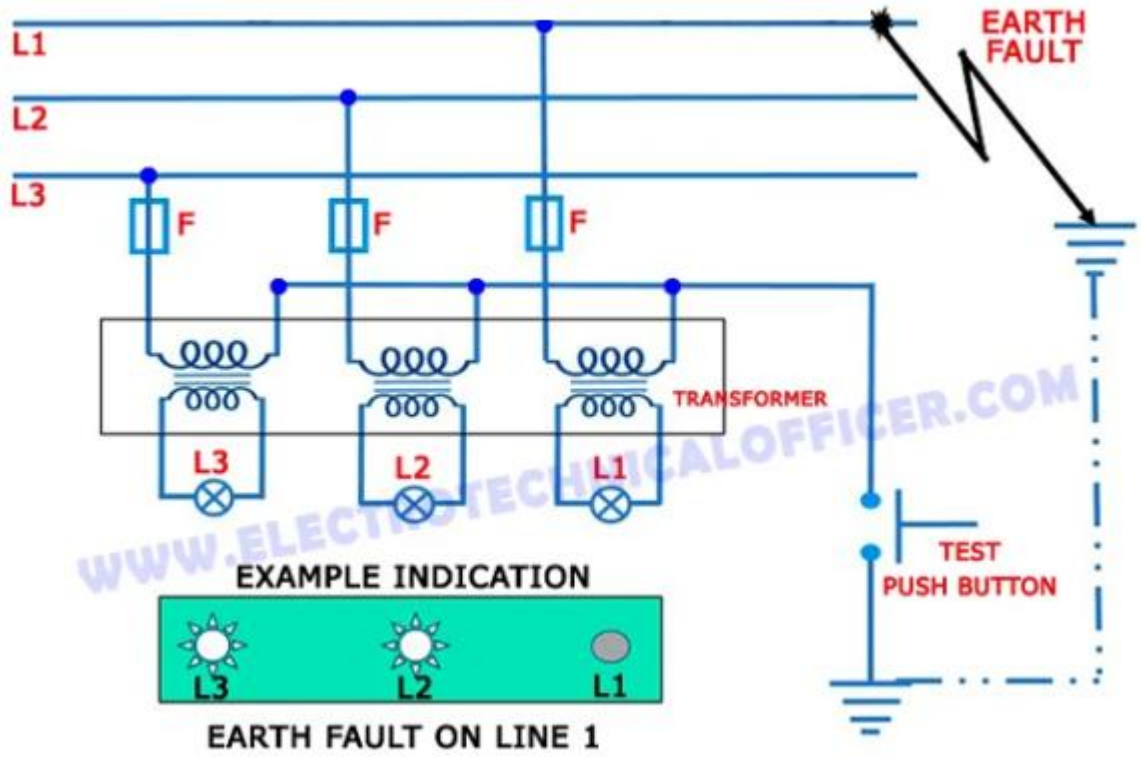


Fig. 3.1: An earth fault indicator

When a connected system is healthy (no earth faults) the lamps glow with equal brightness. However, when an earth fault occurs on one line, the lamp connected to it dims or goes off completely, thus indicating a fault on that line. The disadvantage of this system is that it is not sensitive to high resistance/impedance faults.

The following are common locations of earth faults in ships:

(a) Lamp fittings on the open deck.
(b) At the laundry, washing machines drenched with water.
(c) At the galley, the ovens, hot plates, etc.

(d) Dripping of water over electrical machinery.
(e) Condensed moisture in the motor terminal box.
(f) Overheated motors leading to the melting of varnish on the windings.
(g) Dirty electrical apparatus with surface tracking (leakage current)
(h) Terminal lock nuts becoming loose due to vibration etc., leading to cable touching the body.

Unit 4
MOTOR AND CONTROL EQUIPMENT

4.1 MOTORS

Electric motors are employed in a wide range of applications, in homes and industries. Motors are the most used device for converting electrical energy to mechanical energy in ships. For example, motors are used for electric propulsion, thrusters for propulsion on ship or station keeping, and other on-board loads such as ships winches, pumps, and fans. Typically, 80-90 % of the loads in ship installations will be motor related loads.

There are a variety of electric motors. The selection of these motors is largely based on the nature of the application and the supply available. Every motor has two essential windings, namely, field winding and armature winding. The principal function of field winding is to produce the needed magnetic field. The armature winding uses the magnetic field to generate an adequate torque to make the motor shaft turn.

4.2 TYPES OF ELECTRIC MOTORS

Electric motors can be put into three categories. They are: AC motors, DC motors, and special purpose motors. Figure 4.1 illustrates the types of motors.

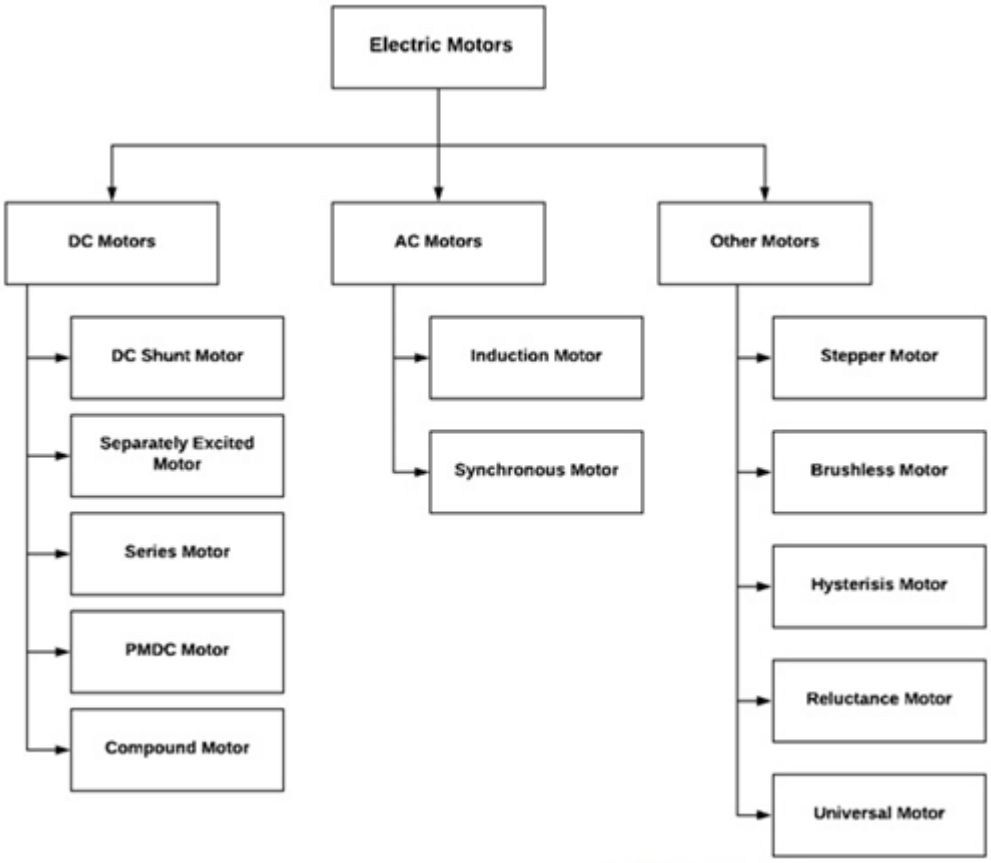


Fig. 4.1 Classes of motors

4.3 **AC MOTORS**

AC motors are classified into synchronous and induction (asynchronous) motors. Figure 4.3 shows the constructional features of AC motor.

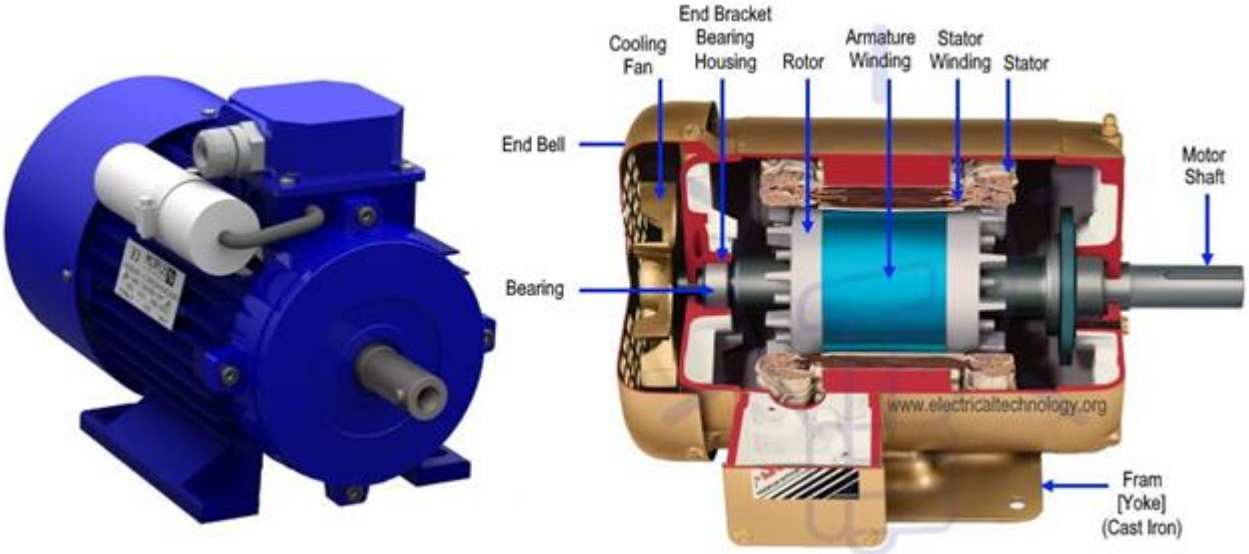


Fig. 4.3: Constructional features of AC motor

4.3.1 **Induction motor**

The asynchronous or induction motor on ship is the workhorse of the industry. Its rugged and simple design ensures in most cases a long lifetime operation, with minimum breakdown and maintenance. The asynchronous motor is used in many applications, either as a constant speed motor directly connected to a network, or as a variable speed motor fed from a static frequency converter. The popularity of 3 phase induction motors onboard ships is because of their simple, robust construction, and high-reliability in the sea environment. An induction motor can be used for different applications with various speed and load requirements. Supplies to induction motors are done via starters, to improve torque and current characteristics.

Based on nature of supply, there are basically two types of induction motors, namely, single-phase and three-phase induction motor. The final output of a three-phase motor is nearly 1.5 times the output (rating) of a single-phase motor of the same size.

(a) ***Single-phase induction motor***

Single-phase induction motors are powered by single-phase power supply. AC current is carried in the main winding of the motor. Single-phase induction motors are mostly used in low power applications such as small pumps, small portable compressors, small fans, and drilling machines.

(b) ***Three-phase induction motor***

Three-phase induction motors are supplied with 3 three-phase AC supply and are widely used in ships for heavier loads. They are of two types, namely, squirrel cage and wound rotor. Squirrel cage

motors are widely used on ships due to their rugged construction and simple design. Typical applications of three-phase induction motors are: lifts, cranes, large capacity exhaust fans, engine auxiliary pumps, engine blower fan motor, engine room heavy load pumps including pumps for ballast, fire-fighting, freshwater and sea water.

Induction motors have the following advantages and disadvantages:
Advantages

- They are robust and simple in construction, with few moving parts.
- They can efficiently operate in a rugged and harsh environment such as in seagoing vessels.
- The maintenance cost of 3-phase induction motor is less and unlike that of DC or synchro motor, they do not have parts like brushes, commuters or slip rings.
- An induction motor can operate in an intrinsic environment as they do not have brushes which can cause sparks that could ignite fire.
- 3-phase induction motors do not need additional starting mechanism or arrangement as they can generate self-starting torque, unlike synchronous motors.

Disadvantages

- During starting, they draw high starting current, particularly when attached to a heavy load. This causes a dip in voltage during the starting period of the machine. Various starting methods are used to avoid this problem.
- Induction motors operate at lagging power factor which results in increased I^2R losses and efficiency reduction, especially at low load. To correct and improve the power factor, static capacitor banks can be used with this type of AC motor.

(c) ***Problems associated with three-phase Induction motor***
Like any other machinery, a three-phase induction motor can face various types of challenges. These can be broadly classified as environmental related faults, electrical related faults and mechanical related faults.

(i) *Environmental related faults*: The harsh environment of the sea may take a toll on the ship's machinery at an early stage if they are not properly maintained. The ambient temperature and the moisture content in the air at sea will affect the operational performance of the induction motor. The motors are installed on other big machinery (main engine) having their own vibrating frequency, which affects the motor parts. A wrong installation or loose foundation of the motor or the load it is connected to may also lead to the reduction in the motor efficiency and eventual failure of the motor.

(ii) *Electrical-related faults*: Problems occur in motors due to internal short circuits and supply related disturbances such as unbalanced supply of voltage. The Different types of electrical faults are:
Winding faults: These result from insulation failures and can lead to serious motor damages.
Single phasing fault: When one or more phases of the 3-phase supply is lost in the course of motor operation, a 3-phase motor will continue to run, but at elevated temperatures and losses.

Crawling: This is when an induction motor runs at extremely low speed (nearly 1/7th of its synchronous speed) due to, for example, the supply having high harmonic content.

(iii) *Mechanical-related faults*: Motors consist of several mechanical parts. The alignment of these parts with each other plays an essential role in motor efficiency. Some of the prominent mechanical related motor faults are:

Imbalance Rotor: The rotor is the only moving part in a 3-phase induction motor. If there is an imbalance in between the shaft rotational axis and weight distribution axis of the rotor, it will create vibration, additional heat and losses in the motor. The unbalance can be due to defect in the rotor, internal misalignment, bending of the shaft, uneven loading and improper motor and load coupling.

Fatigue Failure: If proper maintenance schedules are not followed or parts used in motors are of poor quality, weakening of material may lead to fatigue failure which is usually caused by repeatedly applied loads.

Bearing failure: Motors are fitted with bearing at both ends of the rotor to enable smooth rotation of shaft. The bearings can fail if timely maintenance is not done or due to overloading, wrong installation, contaminated lube oil and operating at excessive temperature. Air or water present in the lubrication (grease) will corrode the bearings, motor shaft and rotors. Motor insulation can also be affected by corrosion, and lead to short circuit between windings.

Lubrication problem: Lack of lubrication or contamination of lubricants can lead to increase friction between parts and bearings may wear out quickly.

(d) **Protections of 3-phase induction motor**

(i) *Overcurrent*

Overcurrent protection interrupts the electrical circuit to the motor upon excessive current demand on the supply system as a result of either short circuits or ground faults in the motor. Overcurrent protection is required to protect personnel, the motor branch circuit conductors, control equipment, and motor from these high currents. Overcurrent protection is usually provided by overcurrent relays (together with circuit breakers) and fuses. These devices operate when a short circuit, ground fault or an extremely heavy overload occurs. Most overcurrent sources produce extremely large currents very quickly.

(ii) *Overload*

One of the most important relays for the detection of abnormal conditions is the overload relay which is applied to the protection of motors. Overload is a condition caused by an increase in the mechanical load to the motor. The overload relay differs from the overcurrent relay in the following ways. Whereas overcurrent relay must operate quickly in times of around or less than 1s, the overload relay is associated with times of tens of seconds to several minutes. However, the overload relay must be capable of measuring accurately current which is only slightly greater than the nominal full-load current, compared to the overcurrent relay, which under fault conditions is required to detect a current which is many times the normal current.

The function of the overload relay is to prevent the overheating of motor. The operation of a typical motor overload relay is of the order of two minutes at twice full-load current. This long time delay can be achieved using thermal relays. Thermal operation is based on the deflection of a bimetallic strip which is made from two metals with differing rates of linear expansion with heat. The circuit is arranged so that the current heats the bimetallic strip which deflects and after a time closes a contact.

(iii) *Phase unbalance*

For three-phase motors, unbalanced phase currents result in a negative phase sequence component which produces a rotating field in the opposite direction to the rotating field produced by the applied system voltage. This counter-rotating field will cause induced currents in the rotor of almost twice normal system frequency, resulting in overheating and possible damage.

Apart from the conditions where one phase of the supply is missing completely, for example, owing to a blown fuse it may be thought that any unbalance in the system is small. This may be true in terms of voltage but as the negative sequence voltage will be applied to the short-circuit impedance of the motor, the current will be substantial. Other conditions which cause unbalanced voltages are heavy single-phase loading or a blown fuse in a power factor correction capacitor circuit. Unbalance relays are employed to curtail the problem associated with phase unbalance.

(iv) *Stalling*

Stalling of the motor refers to a condition, where the motor is unable to rotate. This condition can be caused either due to any obstruction in the load or due to any problem with the motor such as bearing seizure, etc. This condition is also known as locked rotor. When a motor stalls, its slip increases. This causes higher voltage and consequently higher current to be induced in the rotor windings. The stator currents also increase. The equivalent of a motor stalled condition is that of a transformer whose secondary is short circuited. The high current drawn will cause damage to the windings and cause the rotor to heat up. Stall protection devices work by monitoring the motor current and the speed. If the motor draws higher current at a preset low speed, the relay is activated.

The starting current of a direct-on-line motor is practically constant at short-circuit level during most of the run-up period and there is therefore no means of detecting a stalled condition by current level alone. The thermal relay will trip the motor eventually, but because the time is long, it may be too slow to prevent damage. In this case, a single element stalling relay is used. This relay has a directly heated bimetal which gives a low overrun and therefore the operating time can be set close to the maximum run-up time. In some cases, the run-up time is greater than the allowable stall time – which means that the condition can only be resolved by the addition of a speed-measuring relay.

(v) *Insulation failure*

In the great majority of cases, motors are switched by means of contactors; air-break contactors at 415V and vacuum contactors at 3.3kV. At 11kV and sometimes at 6.6kV motors are switched by circuit-breakers. Circuit breakers make and break fault current

whereas contactors make but not break fault current. This means that tripping by a contactor must only be undertaken when the current is less than the contactor capability. That is when the condition detected is overload, unbalance or stalling. Fault current caused by insulation failure must not be cleared by a contactor. The usual method is to clear the fault with fuses which protect not only the motor and the cable but also the contactor. An earth-fault relay can be used as it will detect low level earth faults and trip the contactor, but at high fault levels, the fuse would operate before the relay. Sometimes the earth-fault relay has a time delay to ensure that the fuse is faster.

(e) **Starting of induction motors**

If an induction motor is directly switched on from the supply, it takes 5 to 10 times its full load current and develops a torque which is 2 to 3 times the full load torque. This large starting current produces a large voltage drop in the line, which may adversely affect the operation of other devices connected to the same line. Hence, it is not advisable to start induction motors of higher ratings (generally above 25kW) directly from the mains supply. The methods of starting induction motors are:

(v) Direct-on-line starting
(vi) Resistance starting
(vii) Autotransformer starting
(viii) Star-delta starting
(ix) Soft starting

(i) *Direct-on-line starting*

Induction motors can be started directly on-line using a DOL starter which generally consists of a contactor and a motor protection equipment such as a circuit breaker. A DOL starter consists of a coil operated contactor which can be controlled by start and stop push buttons. When the start push button is pressed, the contactor gets energized and it closes all the three phases of the motor to the supply phases at a time. The stop push button de-energizes the contactor and disconnects all the three phases to stop the motor. In order to avoid excessive voltage drop in the supply line due to large starting current, a DOL starter is generally used for motors that are rated below 5kW. It is simple and low-cost starter for 3-phase induction motors.

(ii) *Resistance starting*

In this method, variable resistors are connected in series with stator windings.

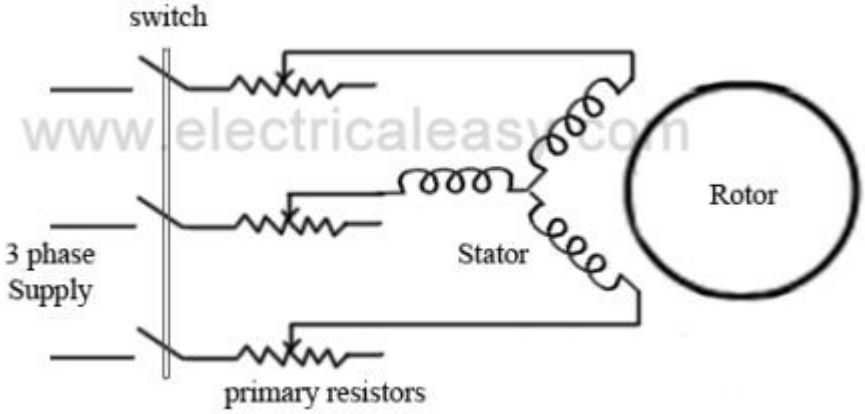


Fig. 4.4: Resistance starter for 3-phase induction motor

The purpose of primary resistors is to drop some voltage and apply a reduced voltage to the stator. Consider, the starting voltage is reduced by 50%. Then according to the Ohm's law (V=I/Z), the starting current will also be reduced by the same percentage. From the torque equation of a three-phase induction motor, the starting torque is approximately proportional to the square of the applied voltage. That means, if the applied voltage is 50% of the rated value, the starting torque will be only 25% of its normal voltage value. Resistors are generally selected so that 70% of the rated voltage can be applied to the motor. At the time of starting, full resistance is connected in series with the stator windings and it is gradually decreased as the motor speeds up. When the motor reaches an appropriate speed, the resistances are disconnected from the circuit and the stator phases are directly connected to the supply lines. This method is generally used for smooth starting of small induction motors. It is however not recommended for starting motors in applications that require high starting torque.

(iii)*Auto-transformer starting*

An auto-transformer is basically a three phase step-down transformer with different taps provided that permit a user to start a motor at, say, 50%, 65% or 80% of line voltage. With auto-transformer starting, the current drawn from supply line is always less than the motor current by an amount equal to the transformation ratio. For example, when a motor is started on a 65% tap, the applied voltage to the motor will be 65% of the line voltage and the applied current will be 65% of the line voltage starting value, while the line current will be 65% of 65% (i.e. 42%) of the line voltage starting value. This difference between the line current and the motor current is due to transformer action. The internal connections of an auto-starter are as shown in figure 4.5. At starting, a reduced voltage (which is selected using a tap) is applied across the stator. When the motor gathers an appropriate speed, say up to 80% of its rated speed, the auto-transformer automatically gets disconnected from the circuit as the switch goes to "run" position.

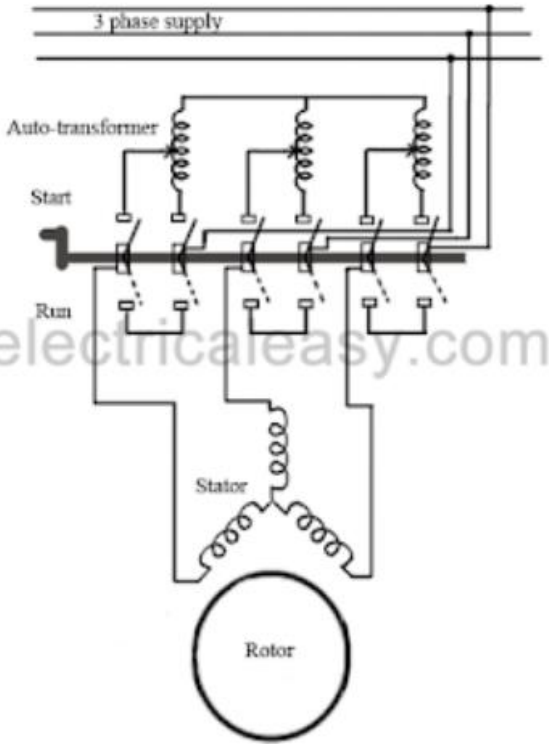


Fig. 4.5: Autotransformer starting

(iv)*Star-Delta Starting*

This method is used in motors that are designed to run on delta connected stator. A two-way switch is used to connect the stator winding in star while starting, and in delta while running at normal speed. When the stator winding is star connected, voltage over each phase in motor will be reduced by a factor 1/(sqrt. 3) of that for delta connected winding. The starting torque will 1/3 times that delta connected winding. Hence a star-delta starter is equivalent to an auto-transformer of ratio 1/(sqrt. 3) or 58% reduced voltage.

4.3.2 **Synchronous motors on Ship**

The synchronous motor is normally not used as a motor in ship applications, with exception of large propulsion ship drives, typically > 5 MW directly connected to propeller shaft, or >8-10MW, with gear connection. In power ranges smaller than these, the asynchronous motor is normally cost-competitive. The design of a synchronous motor on ship is similar to that of a synchronous generator. Synchronous motor is used on ship as prime movers (drives) for centrifugal pumps, belt-driven reciprocating compressors, and many more applications.

(a) *Construction of synchronous motor*

The structure is same as of other motors. Stator and rotor are the main parts of a synchronous motor while a frame is the cover and both stator and rotor make up the electric and magnetic circuitry of synchronous motors. The main components of the synchronous motor are: Stator, Rotor, Exciter, and Frame. Figure 4.6 shows the constructional details of a synchronous motor.

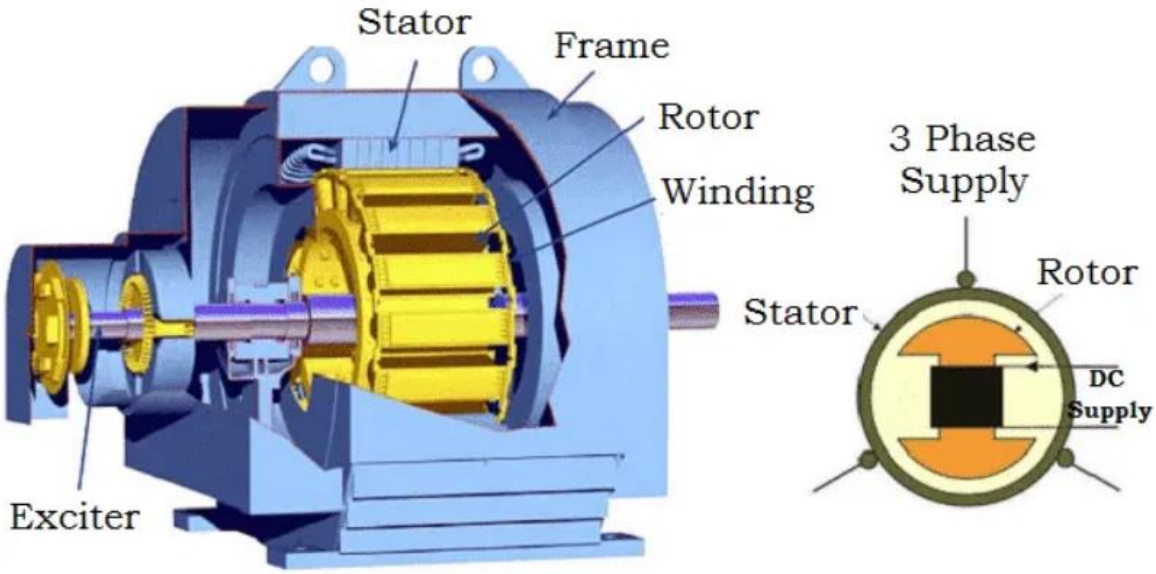


Fig. 4.6: Components of synchronous generator

The rotor winding is excited by a DC source and the stator winding excited by 3-ph AC source. Due to the 3-phase source, a 3 phase rotating magnetic field is produced by stator winding. Rotor winding produces a constant magnetic field. At some rotations, the poles of two magnetic fields attract each other while at some instant, they repel each other. The rotor will not start to rotate due

to its inertia. An external source is used to provide an initial rotation. Once the rotor starts moving at synchronous speed, the external source is cut off.

Advantages of synchronous motors

- It can control power factor. An over excited synchronous motor has leading power factor and is operated in parallel to induction motors thereby improving the system power factor.
- Speed remains constant irrespective of the loads in synchronous motors. This quality helps in industrial machines where constant speed is required irrespective of the load.
- Synchronous motors are built with wider air gaps than induction motors which make them mechanically more stable.
- Electro-magnetic power varies linearly with the voltage in synchronous motors. Synchronous motors usually operate with higher efficiencies (more than 90%) especially in low speed compared to induction motors.

Disadvantages of synchronous motors

- Synchronous motors require dc excitation which is supplied from external sources.
- They are largely not self-starting motors and need some external arrangement for starting and synchronizing.
- The cost per kW output is commonly higher than that of induction motors.
- Unless the incoming supply frequency is adjusted, there is no possible way to adjust the speed.
- They cannot be started on load because its starting torque is zero.
- Collector rings and brushes are required which results in high maintenance cost.
- Synchronous motors cannot be useful for applications requiring frequent starting of machines.

(b) *Starting methods of synchronous motor*

A classical synchronous motors cannot self-start, as it has no starting torque. One method used to make a synchronous motor self-start is the introduction of a squirrel cage arrangement fitted through pole tips. At the starting, rotor field coils are not energized, so with revolving magnetic field electricity is induced in squirrel cage bars and the rotor starts rotating just like an induction motor. When the rotor has achieved its maximum speed, rotor field coils are energized. The poles of rotor get locked with poles of RMF and will start rotating at synchronous speed.

(c) *Permanent magnet synchronous motors on Ship*

Permanent magnet synchronous motors are employed for ship propulsion. The benefit of this design is high efficiency with compact design, making it especially interesting for podded propulsion where the dimensions should be as small as possible, and direct water cooling would eliminate the need for air cooling of the pod motor and simplify the construction and installation work.

4.4 **DC MOTORS**

The types of dc motors include Series, Shunt, Compound wound, and Permanent Magnet DC motors (PMDC) motor. Figure 4.2 shows the general constructional features of a dc motor.

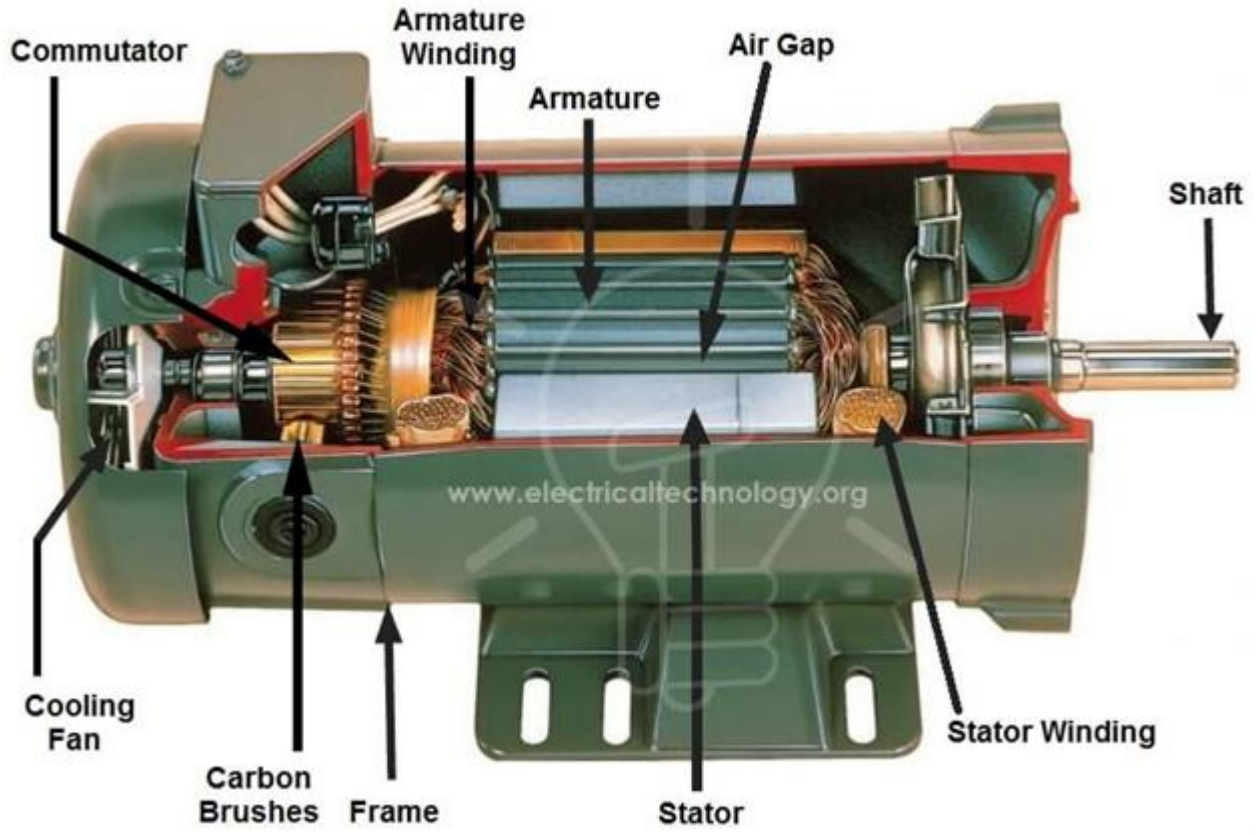


Fig. 4.2: Constructional features of DC motor

(a) *DC Shunt Motor*

This kind of motor is also called as shunt wound DC motor, where the winding type is known as a shunt winding. DC motors are used wherever stable speed is required. They are employed in centrifugal pumps, lifts, weaving machine, lathe machines, blowers, fans, conveyors, spinning machines, etc.

(b) *Separately Excited Motor*

In separately excited motor, the connection of stator and rotor can be done using a different power supply. So that the motor can be controlled from the shunt and the armatures winding can be strengthened to generate flux.

(c) *DC Series Motor*

In DC series motor, rotor windings are connected in series. These motors are mainly used in starter motors which are used in elevators and cars. These motors can produce enormous rotating force and torque, from inactive state. This feature makes the series motor fit for mobile electric equipment, tiny electrical appliances, winches, hoists, etc.

(d) *Compound DC Motor*

Generally, DC compound motor is a hybrid of DC series and shunt motors. It has both series and shunt fields. This combination presents a double advantage of having the torque characteristics of a series motor and the constant speed characteristic of a shunt motor in one compound wound motor. Compound motors are suitable for almost all applications like business machines, machine tools, agitators, and mixers. They are used to drive loads such as shears, presses and reciprocating machines.

(e) *PMDC Motor*

Permanent Magnet DC motors, as the name suggest are built to included permanent magnets to produce the needed magnetic field. These motors are in several low and high power applications. They are used in automobiles for operating windshield wipers as well as washers, to move up and lower windows, to drive blowers for air conditioners as well as heaters. They are used in computer drives, toy industries, food mixers, electric toothbrushes, and moveable vacuum cleaners. They are also used in handy electric tools like hedge trimmers and drilling machines.

4.5 **SPECIAL PURPOSE ELECTRIC MOTOR**

The special purpose motors include stepper motor, brushless dc motor, hysteresis motor, and universal motor.

(a) *Stepper Motor*

The stepper motor can be used to offer step angle revolution, as an alternative to stable revolution. In a stepper motor, the complete revolution angle can be separated in numerous steps like 10 degree X 18 steps. This means, in a total revolution cycle the rotor will go stepwise eighteen times, at 10 degree each time. The applications of stepper motor include the following:

Industrial Machines – Automotive gauges and machine tooling automated production equipment.
Security – Surveillance products for security industry.
Medical – Medical scanners, samplers, digital dental photography, fluid pumps, respirators, and blood analysis machinery.
Consumer Electronics –Aautomatic digital camera focus and zoom functions.

(b) *Brushless DC Motors*

The brushless DC motors were first developed for achieving superior performance within a lesser space than brushed DC motors. A controller is embedded into the electric motor to enable its operation in the absence of a commutator and slip ring.

(c) *Hysteresis Motor*

The operation of the hysteresis motor is particularly unique. These motors give a very smooth operation with stable speed. The noise level of this motor is quite low. Due to this, they are found in applications that require motors to be soundproof such as sound player and audio recorder.

(d) *Universal Motor*

This is a special kind of motor that works when supplied with either AC or DC supply. Universal motors are series wound; with the field and armature windings connected in series. It generates high starting torque. They are mainly designed for operating at high-speeds, above 3500 rpm.

4.6 **SPEED CONTROL OF SHIP MOTORS**

The standard cage-rotor induction motor operates as an almost constant speed drive over its load range. This feature is satisfactory for most of ship's auxiliary services supplying power to ventilation fans and circulating pumps. However, variable speed control is necessary for cranes, winches, windlass, capstans, forced-draught fans, etc. The direct on-line ship motor will rotate with a speed directly determined by the ship network frequency. Common speed controllers are:

(a) *AC controllers:*

An AC controller (Sometimes referred to as a variable frequency drive, adjustable speed drive, frequency converter, or Driver) is a device that controls the speed of an AC motor. The AC Motor receives power, which is ultimately converted by the AC controller into an adjustable frequency. The adjustable output allows the motor speed to be precisely controlled. Typically, an AC controller consists of three basic parts: rectifier, inverter, and DC link to connect the two. The rectifier converts AC input into DC, while the inverter switches the DC voltage to an adjustable frequency AC output voltage. The inverter can also be used to control output current flow if needed. Both the rectifier and inverter are directed by a set of controls to generate a specific amount of AC voltage and frequency to match the AC motor system at a given point in time.

An AC controller can be used in many different industrial and commercial applications. It is often used to control fans in air conditioning and heating systems. The AC controller also aids in adjusting the speed of pumps and blowers. More recent applications include conveyors, cranes and hoists, machine tools, extruders, film lines, and textile-fiber spinning machines.

The offer the following benefits:

- Increases the life of the motor due to high power factor
- Economical speed control
- Optimize motor-starting characteristics
- Lower maintenance than DC control

However, they generate a large amount of heat and harmonics.

(b) *Variable Frequency Drives (VFDs)*

The variable frequency drive is a particular kind of adjustable-speed drive that is used to control the speed of an AC motor. In order to control the motor's rotational speed, a variable frequency drive controls the frequency of the electrical power supplied to it. Adding a variable frequency drive to an application allows the motor speed to be adjusted in accordance with the motor's load,

ultimately saving energy. Commonly used in a myriad of applications, a variable frequency drive can be found operating ventilation systems, pumps, conveyors and machine tool drives.

When complete voltage is applied to an AC motor, it accelerates the load and drops torque initially, keeping current especially high until the motor reaches full speed. A variable frequency drive operates differently; it eliminates excessive current, increasing voltage and frequency in a controlled manner as the motor starts. This allows the AC motor to generate up to 150% of its rated torque, which could potentially be generated from the beginning, all the way up to full speed, without wasting energy. A Variable frequency drive converts power through three different stages. First, AC power is converted to DC power, followed by the switching on and off of power transistors, causing a voltage waveform at the desired frequency. This waveform then adjusts output voltage according to the preferred designated value.

Typically, a variable frequency drive system involves an AC motor, controller, and operator interface. VFD is commonly applied three-phase induction motors they offer versatility and cost-effectiveness compared to a single-phase induction motors or synchronous motors.

The advantages associated with employing VDFs are:

- Process temperature can be controlled without a separate controller
- Low maintenance
- Longer lifespan for the AC motor and other machinery
- Lower operating costs
- Equipment in the system that cannot handle excessive torque is protected

There are three common VFDs that offer both advantages and disadvantages depending on the application they are used for. The three common VFD designs used include: Current Source Inverter (CSI), Voltage Source Inverter (VSI), and Pulse Width Modulation (PWM). However, there is a fourth type of VFD called Flux Vector Drive, which is emerging in popularity among end-users for its closed-loop control feature. Each VFD consists of a Converter, DC Link and Inverter section but how each one is constructed varies from drive to drive.

(i) *Current Source Inverter (CSI)*

A Current Source Inverter (CSI) is a type of variable frequency drive (VFD) which converts incoming AC voltage and varies the frequency and voltage supplied to the associated motor. The general configuration of this type of VFD is like that of other VFDs in that it consists of a converter, DC Link, and inverter. The converter part of the CSI uses silicon-controlled rectifiers (SCRs), gate-commutated thyristors (GCTs) or symmetrical gate-commutated thyristors (SGCTs) to convert the incoming AC voltage to a variable DC voltage. To maintain the correct voltage to frequency (Volt/Hertz), the voltage must be regulated by the correct sequencing of the SCRs. The DC Link for this type of variable frequency drive uses an inductor to regulate the current ripple and to store the energy used by the motor. The inverter, which is responsible for converting the DC Voltage back to an AC sine-like waveform, comprises of SCRS, gate turn-off thyristors (GTOs) or symmetrical gate-commutated thyristors (SGCTs). These thyristors behave like switches which are turned on and off to create pulse width modulation (PWM) output that regulates the frequency and voltage to the motor. CSI variable frequency drives regulate current, require a large internal inductor and a motor load to operate. An important note about CSI VFD designs is the requirement

of input and output filters which are necessary due to high harmonics in the power input and poor power factor. To work around this issue, many manufacturers implement either input transformers or reactors and harmonic filters at the point of common coupling (users electrical system connected to the drive) to help reduce the effects harmonics have on the drive system. Of the common VFD drive systems, CSI VFDs are the only type of drives that have regenerative power capability. Regenerative power capability means that power is driven back from the motor to the power supply. The advantages of CSI are:

- Regenerative power capability
- Simple circuitry
- Reliability (Current Limiting Operation)
- Clean current waveform

The disadvantages of CSI are:

- Motor cogging when PWM output is below 6Hz
- Inductor used are large and costly
- Large power harmonic generation sent back into power source
- Dependent on motor load
- Low input power factor

(ii) **Voltage Source Inverter (VSI)**

The converter section of the VSI is similar to the converter section of the CSI in that the incoming AC Voltage is converted into a DC Voltage. The difference between the CSI and VSI converter section is that the VSI uses a diode bridge rectifier to convert the AC Voltage to DC Voltage. The DC Link of the VSI uses capacitors to smooth out the ripple in the DC voltage and to also store energy for the drive system. The inverter section is comprised of insulated gate bipolar transistors (IGBTs), insulated gate-commutated thyristors (IGCTs) or injection-enhanced gate transistors (IEGTs). These transistors or thyristors behave like switches which are turned on and off to create a pulse width modulation (PWM) output that regulates the frequency and voltage to the motor. The a**dvantages of VSI are:**

- Simple Circuitry
- Can be used with applications requiring multiple motors
- Not dependent on load

The disadvantages of VSI are:

- Large power harmonic generation into power source
- Motor cogging when PWM output is below 6Hz
- Non-Regenerative operation
- Low power factor

(iii) **Pulse Width Modulation (PWM)**

The Pulse Width Modulation (PWM) Variable Frequency Drive (VFD) is among the commonly used controllers and has proven to work well with motors that range in size from 1/2HP to 500HP. Most PWM VFDs are rated for 230V or 460V, 3-phase operation, and provide output frequencies in the range of 2-400Hz. Like the VSI VFD, the PWM VFD uses a diode bridge rectifier to convert the incoming AC voltage to a DC voltage. The DC Link uses large capacitors to remove the ripple

evident after the rectifier and creates a stable DC bus voltage. The six-step inverter stage of this driver uses high power rated IGBTs which turn on and off to regulate the frequency and voltage to the motor. These transistors are controlled by a microprocessor or motor IC which monitors various aspects of the drive to provide the correct sequencing. This produces a sine-like waveform output to the motor. The frequency supplied to the motor is determined by the number of positive to negative transitions per second. The advantages are as follows:

- No motor cogging
- Efficiencies from 92% to 96%
- Excellent input power factor due to fixed DC bus voltage
- Low initial cost
- Can be used with applications requiring multiple motors

The disadvantages are:

- Non-Regenerative operation
- High frequency switching may cause motor heating and insulation breakdown

(c) *Cycloconverter*

Cycloconverters are frequency changers that convert AC power of specific frequency and voltage to different frequency and voltage of AC power without any intermediate DC link. A cycloconverter converts the mains frequency into a variable frequency directly through a one-step conversion process. This is the essential difference between it and the other converters. A cycloconverter can be programmed to generate variable-frequency variable-voltage to drive a motor.

Vessel motor controllers acquire measurement signals and feedback signals from sensors in the drive, and motor. Typically, ship motor currents, motor speed, and in some cases temperatures and voltages are measured.

UNIT 5
OTHER MARINE ELECTRICAL EQUIPMENT

5.1 **BOILERS**

A boiler is one of those machineries that gets ships going. A boiler is an equipment, which though not required continuously in operating a ship, cannot be done away with. Boilers generate steam at extremely high pressure. Therefore, proper care should be taken while operating them.

When a boiler is started, its lighten-up rate must not be accelerated too much as this might cause an unnecessary strain of the boiler material by quick and uneven temperature rises. It might be necessary to perform several start/stop sequences to reduce the lighten-up rate. At the commissioning start-up of a boiler and after any repair work of a refractory, it is important to further reduce the lighting-up rate. This is because the new refractory still contains a small amount of water. When heated, the water vaporises and expands which might cause fissures and cracks in the refractory. Burners must therefore only be operated at minimum load and in intervals of 1-2 minutes for the first hours. Between each operation interval, the burner should remain stopped for approximately 8-10 minutes.

A step-by-step procedure is outlined for starting and stopping a boiler during normal and emergency working conditions.

5.1.1 **Precautions on Starting Boiler**

Before start-up of the boiler plant, the following general work and check procedures must be considered.

1. Check that the main steam valve, by-pass valve and circulation valves, if provided, scum valve, and blow-down valves are closed.
2. Open feed water valves and air valve. Fill the boiler with feed water to approximately 50mm below normal water level. The water level rises due to expansion when the boiler is heated. If the temperature difference between the boiler and feed water exceeds approximately 50ºC, the boiler must be filled slowly.
3. When filling a pressureless boiler, the shut-off valve after the feed water pump must be throttled. Otherwise, the pump motor will be overloaded.
4. Check the water level in the water level gauges. Check frequently during the complete start-up. The water level gauges should be blown down several times to ensure a correct indication.
5. Check that the water level control system is connected and operational.
6. Check the oil system and start the fuel oil supply pump. Pre-heat the fuel oil if the burner should operate on heavy fuel oil.
7. Check the burner and the safety functions according to the manufacturer instruction.

5.1.2 **Starting Boiler and Pressure Rise**

The following general work procedures must be followed during start-up of aboiler.

1. Check that the gauge board valve and pressure gauge valves are opened.
2. Check that the air valve is open if the boiler pressure is below 1.0 bar.
3. Start the burner on manual control and on low load. Check that the water level does not rise too high during the pressure rising period.

4. Drain via the blow down valves if the water level is too high.
5. If the air valve was opened close it when only steam blows out. A pressure reading should be indicated on the boiler pressure gauge before the air valve is closed.
6. Tighten all covers such as manholes, hand holes, inspection doors, etc. during the pressure rising period. If required, check all flange joints on the plant.
7. Change to automatic control of the burner when the boiler pressure is 0.5 bar lower than the working pressure of the boiler.
8. Open the by-pass valve slowly to heat-up and pressurise the steam system. If the boiler is not provided with a by-pass valve, the main steam valve should be used to heat-up and pressurise the steam system.
9. Open the main steam valve and close the by-pass valve.
10. Open the valves to the steam consumers carefully to avoid water chocks.
11. When the boiler is in normal operation, check that the water level control system and the gauge board functions are fully operational.

After 3-4 weeks in operation, mud and deposits in the piping system may have accumulated in the boiler water. This may cause level variations which disturb steam generation. It is therefore recommended to blow down the boiler. It should then be inspected, cleaned, and refilled with boiler water.

5.1.3 **Stopping boiler**

Normal boiler shutdown

When necessary, a boiler can be shut down at any load without special preparations. When a boiler is stopped, sudden temperature and pressure drops should be avoided as they might expose mountings, pipelines, and boiler plant to inadmissible temperature gradients. The following can be done:

1. When minimum load is obtained, stop the burner.
2. Keep water level at normal level until the boiler stops producing steam.
3. Stop feed water pump and close the feed water valves.
4. Close main steam valve.

Emergency shutdown

A boiler must be taken out of service immediately, if parts of the heating surface glows or the boiler shows recognisable deformations. The supervising authorities must be informed, and the boiler must not be used until approval from these authorities is available. If emergency shutdown must be carried out, the fuel supply should be stopped. The main steam valve should be closed gradually, and the boiler must be cooled. The safety valves must not be operated. Parallel working boilers should be disconnected at once.

Stopping for repair or inspection

The following describes the measures to be taken when a boiler is shut down for repair or inspection.

1. Clean the boiler from soot, with water.
2. Operate the burner for at least 15 minutes after the soot removal to dry out the remaining water.
3. Stop the boiler as mentioned previously.

4. Check the furnace and the pin tubes with regard to cleanliness.
5. Empty the boiler from water and clean it. Check if lime stone appears.
6. Check and clean the outer fittings. Change gaskets where required.
7. Clean the feed water tank and feed water pipes.
8. Clean and grease the bearings of motor, pump, and fan.
9. Check and align the burner, if necessary.
10. If the boiler is shut down for a long period of time, the pin tubes must be thoroughly cleaned.
11. Check that the necessary spare parts are available. Order complementary parts in time.

It is of extreme importance that a boiler is NOT operated without water, for example, due to disconnection of the water level safety devices when the oil burner is in operation. This will immediately cause complete breakdown of the boiler.

5.2 **REFRIGERATION PLANT**

Refrigeration is a process by which the temperature of a space or its contents is reduced to levels below that of their surroundings.

The refrigeration plants on merchant vessels play a vital part in carrying refrigerated cargo and provisions for the crew on board. In reefer ships, the temperature of perishable or temperature sensitive cargo such as food, chemical, or liquefied gas, is controlled by refrigeration plants. The same plant or a smaller unit can be used for maintaining the temperature of different provision rooms carrying food stuffs for crew members. The main purpose of ship's refrigeration plant is to avoid any damage to cargo or perishable materials and enable transportation in good and healthy condition. Refrigeration prevents growth of micro-organisms, oxidation, fermentation and drying out of cargo, etc. Refrigeration is also required to cool bulk CO_2 for fire-fighting systems.

Ships refrigeration plant may vary from small domestic refrigerating unit, for provisions, to large plants, for reefer vessels. There are four basic components of a refrigeration system. These are:

(a) Compressor
(b) Condenser
(c) Expansion valve
(d) Evaporator.

Compressor: The function of the compressor in a refrigeration system is to raise the pressure of the vapourised refrigerant, causing its saturation temperature to rise so that it is higher than that of seawater or an air cooled condenser. The compressor also promotes circulation of the refrigerant by pumping it around the refrigeration system.

Condenser: The function of the condenser is to liquefy the refrigerant and sub cool it to below the saturation temperature by circulating seawater or air. Latent heat originally from the evaporator is transferred to the cooling medium. The liquid refrigerant still at pressure produced by the compressor passes on to the expansion valve.

Expansion valve: The function of the expansion valve in a refrigeration system is to regulate the flow of refrigerant from the HP side of the system to the LP side of the system. A drop in pressure

causes the saturation temperature of refrigerant to fall so that it will boil at low temperature, in the evaporator. The expansion valve controls the flow of refrigerant to the evaporator, thermostatically.

Evaporator: The function of the evaporator in a refrigeration system is to cool the air in the fridge space. It does this because the temperature of the refrigerant entering the evaporator is lower than that of the air in the space and this causes the refrigerant to receive latent heat and evaporate. The evaporator normally has a fan to circulate the air around it.

Refrigeration of cargo spaces and storerooms employs a system of components to remove heat from the space being cooled. This heat is transferred to another body at a lower temperature. The cooling of air for air conditioning entails a similar process. The transfer of heat takes place in a simple system: firstly, in the evaporator where the lower temperature of the refrigerant cools the body of the matter being cooled; and secondly, in the condenser where the refrigerant is cooled by air or water. The usual system employed for marine refrigeration plants is the vapour compression cycle as shown in figure 5.1.

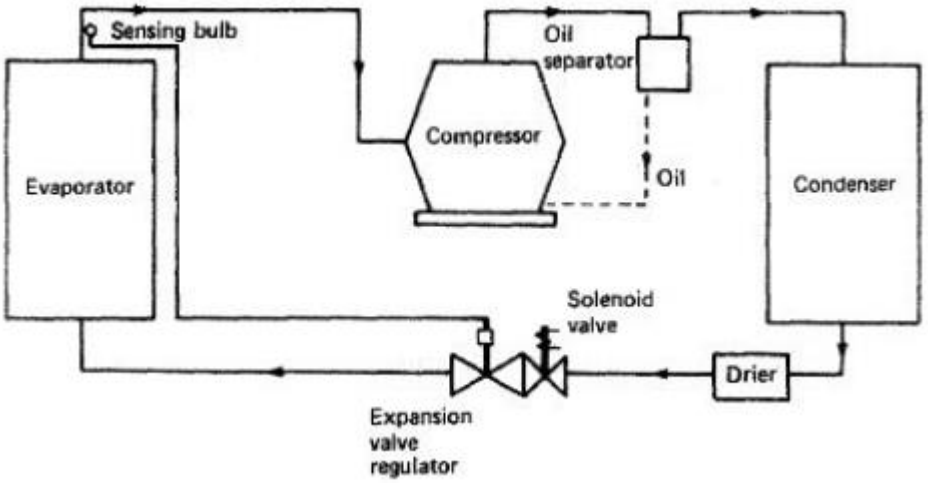


Figure 5.1: Vapour compression cycle

The pressure of the refrigerant gas is increased in the compressor and it thereby becomes hot. This hot, high-pressure gas is passed through into a condenser. Depending on the particular application, the refrigerant gas will be cooled either by air or water, and because it is still at a high pressure it will condense. The liquid refrigerant is then distributed through a pipe network until it reaches a control valve alongside an evaporator where the cooling is required. This regulating valve meters the flow of liquid refrigerant into the evaporator, which is at a lower pressure. Air from the cooled space or air conditioning system is passed over the evaporator and boils off the liquid refrigerant, at the same time cooling the air. The design of the system and evaporator should be such that all the liquid refrigerant is boiled off and the gas slightly superheated before it returns to the compressor at a low pressure to be recompressed. Thus, it will be seen that heat that is transferred from the air to the evaporator is then pumped round the system until it reaches the condenser where it is transferred or rejected to the ambient air or water. It should be noted that where an air-cooled condenser is employed in small plants, such as provision storerooms, adequate ventilation is required to help remove the heat rejected by the condenser. Also, in the case of water-cooled condensers, fresh water or sea water may be employed. Fresh water is usual when a central fresh-

water/sea-water heat exchanger is employed for all engine room requirements. Where this is the case, because of the higher cooling-water temperature to the condenser, delivery temperatures from condensers will be higher than that on a sea water cooling system.

5.2.1 Safety devices on refrigeration plant

To have an efficient and hassle free operation of a ship's refrigeration plant and its components, a good maintenance schedule and safety systems are adopted. A safety system includes alarm, cut offs, and trips which safeguards the machinery and its parts from getting damage. The main safeties adopted for refrigeration plants are:

***Low Pressure or LP cut off*:** This is a compressor safety feature that cuts off the compressor in the event of pressure drop in the suction line. The pressure of the suction line is continuously sensed by the control unit and when it goes below the set value, which means the room is properly cooled, the LP cut out will auto trip the compressor. When the pressure rises, indicating there is flow of refrigerant in the line due to increase in room temperature, the LP switch will start the compressor.

***High pressure or HP cut out*:** As the name suggests, the high pressure cut out activates and trips the compressor when the discharge side pressure increases above the limit value. This can lead to overloading of compressor parts and may damage same. The HP cut out is not auto reset and has to be done manually.

***Oil differential cut out*:** This safety measure is again applied to compressors as they are the only machinery in refrigeration systems that have rotational parts and hence require continuous lubrication. In the event of low supply or no supply of lube oil to the bearing, the differential pressure will increase and activates a trip signal to safeguard bearing and crankshaft.

***Relief valves*:** Relief valves are fitted on the discharge sides of compressors. They safeguard compressors in the event of over pressure. One relief valve is also fitted in condenser refrigerant line to avoid damage to the condenser if there is high pressure in the discharge line.

***Solenoid valves*:** A master solenoid is fitted after the receiver, which is controlled by the control unit. In case of sudden stoppage of compressor, the master solenoid also closes, avoiding the flooding of evaporator with refrigerant liquid. The room or hold solenoid and thermostatic valve regulate the flow of the refrigerant into the room to maintain the temperature of the room.

***Oil heater*:** Oil heater is provided for compressor crank case oil and prevents compressor from getting excessively cold, which may affect the lubrication of parts.